Adaptive Web Design and Content Delivery using Multi-Modal Generative AI: Personalizing Experience through User Characteristics

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Authorship Statement

This dissertation is based on the results of research carried out by myself, is my own composition, and has not been previously presented for any other certified or uncertified qualification.

The research was carried out under the supervision of Mr.Daren Scerri.

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I would like to express my heartfelt gratitude to my mentor for their guidance, support, and valuable insights throughout the course of this dissertation. I am also deeply thankful to my family and friends for their continuous encouragement and understanding during this journey. Lastly, I give thanks to God for granting me the strength, patience, and clarity to complete this work.

Abstract

This study explores the use of Large Language Models (LLMs) to develop a framework for adaptive web page design and content delivery. For the purpose of this experiment, personalisation of health information was chosen as the applied area to test the solution, through user characteristics variables of age and gender. A prototype system was developed using DeepSeek r1:7b for text generation and BLIP for image-caption matching, enabling the automatic creation of user-specific health advice webpages. The system was evaluated through two approaches: system-level testing assessed performance consistency and functional reliability across multiple runs, while user testing compared the adaptive prototype with a static, non-personalised version. Ten participants interacted with both versions and provided feedback on relevance, usability, and satisfaction. Results showed that the personalised prototype was rated higher in terms of content relevance and user engagement, with most participants affirming that personalisation improved their experience. However, issues such as occasional image mismatches and layout preferences were identified. Overall, the findings suggest that LLM-driven personalisation enhances web-based health communication, though further development is required to address usability and deployment limitations.

Keywords: Adaptive Web Design, Large Language Models, Personalised Content Delivery, User-Centred Design, Vision-Language Models, Usability Evaluation.

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List of Abbreviations

AI	Artificial Intelligence
HCI	Human-Computer Interaction
LLM	Large Language Model
UCD	User-Centred Design
BLIP	Bootstrapped Language Image Pretraining
API	Application Programming Interface
AR	Augmented Reality
VR	Virtual Reality
CDN	Content Delivery Network
GDPR	General Data Protection Regulation

Chapter 1: Introduction

1.1 Background and Context

Web-based content delivery has advanced considerably in recent years, with modern systems expected to respond dynamically to user characteristics, behaviours, and needs [5]. This growing sophistication has led to the integration of intelligent systems capable of tailoring not just layout and structure, but also the substance of content in real time. Despite these advancements, a considerable portion of websites—particularly those delivering health-related information—continue to rely on static, one-size-fits-all content that fails to account for fundamental user characteristics such as age or gender.

At the same time, the broader technological landscape has witnessed the widespread adoption of AI-driven interfaces in consumer-facing domains. Chatbots, for example, are now being used across healthcare, customer service, and education to deliver information in a conversational and adaptive manner [6]. These systems leverage natural language processing to offer more interactive and personalised user experiences. As expectations rise for systems that can "understand" and respond to individual needs, the demand for more adaptive and intelligent content delivery methods continues to grow.

This dissertation will explore how LLMs can be applied to the challenge of personalised health content delivery. Specifically, it will investigate whether a prototype system that adapts its output based on user characteristics such as age and gender can deliver more relevant and engaging experiences than static, generic alternatives.

1.2 Research Problem

The central problem this research aims to address is whether a content delivery system powered by LLMs can meaningfully improve user engagement, satisfaction, and perceived relevance when compared to traditional static health content. In contexts such as healthcare, where clarity, trust, and user-specific applicability are vital, the ability to personalise web content could enhance user outcomes and increase the usefulness of digital resources.

1.3 Rationale for the Study

The motivation for this study lies in the ongoing shift from generic to personalised user experiences across digital platforms. As users increasingly seek content that reflects their unique needs and situations, conventional approaches to information delivery have begun to show their limitations. This is particularly evident in the healthcare sector, where users benefit greatly from advice that is directly applicable to their age, gender, or lifestyle. The emergence of powerful LLMs and vision-language models offers new ways to meet these demands by generating content that feels tailored, inclusive, and contextually relevant.

1.4 Research Aims

The aim of this study is to investigate whether LLMs can be effectively used to generate personalised web-based health content that enhances relevance and user engagement compared to static, non-adaptive approaches.

This study will follow a design–build–evaluate methodology to explore the potential of LLMs in adaptive content generation. A working prototype will be developed that generates a personalised webpage containing health tips and matched visual content, based solely on simple user inputs. The system will combine a vision-language model for selecting appropriate imagery and an LLM for generating and formatting the text into HTML. The prototype will be evaluated through both system-level testing—focusing on consistency, generation time, and visual coherence—and user-based testing involving ten participants who will interact with both the personalised and static versions.

1.5 Scope and Significance

This research will focus on the development and evaluation of a proof-of-concept system that uses LLMs to deliver personalised health content via the web. It will address a key shortcoming in current web-based delivery systems: the continued reliance on static, one-size-fits-all formats, even in sectors like healthcare where relevance and clarity are vital.

A comparison of the NHS website over an eight-year span reveals this limitation clearly, using snapshots from the Wayback Machine[1]. Figure 1.1 shows the NHS homepage as it appeared in 2015. The design features a tiled layout, drop-down navigation, and a fixed set of content categories such as "Health A-Z" and "Services Near You." This layout, while functional, offered no dynamic content or adaptation based on user needs or characteristics.

In Figure 1.2, the 2023 version of the NHS site presents a cleaner visual interface with modernised typography and spacing. However, the core structure remains largely unchanged. The site still delivers content through static templates and uniform navigation paths, with no evidence of personalised health recommendations or adaptive interfaces based on demographic inputs. This reflects a broader trend in public service platforms where design improvements focus on aesthetics rather than adaptive functionality.

This dissertation aims to demonstrate that systems powered by LLMs can go beyond surface-level redesigns by offering dynamic, user-aware content generation. The significance of this lies not just in improved engagement, but also in increasing the accessibility and usefulness of health information for diverse audiences. Through the development and evaluation of a working prototype, this study

[1]Wayback Machine: https://web.archive.org/

contributes to the growing field of intelligent web interfaces capable of adapting in real time to user characteristics such as age and gender.

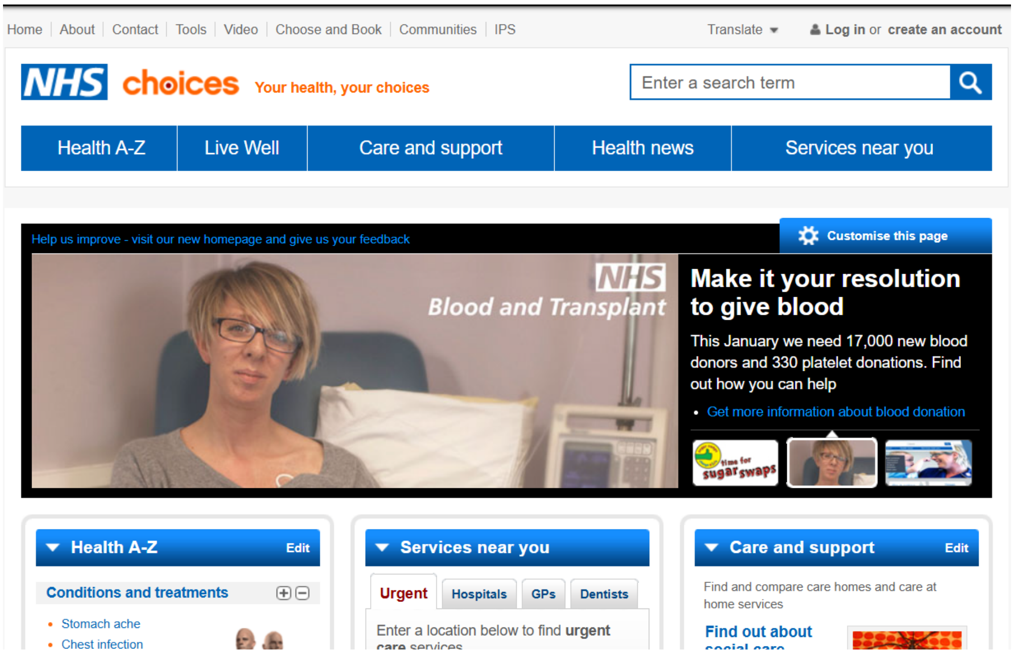

Figure 1.1: *NHS website layout from 2015.*

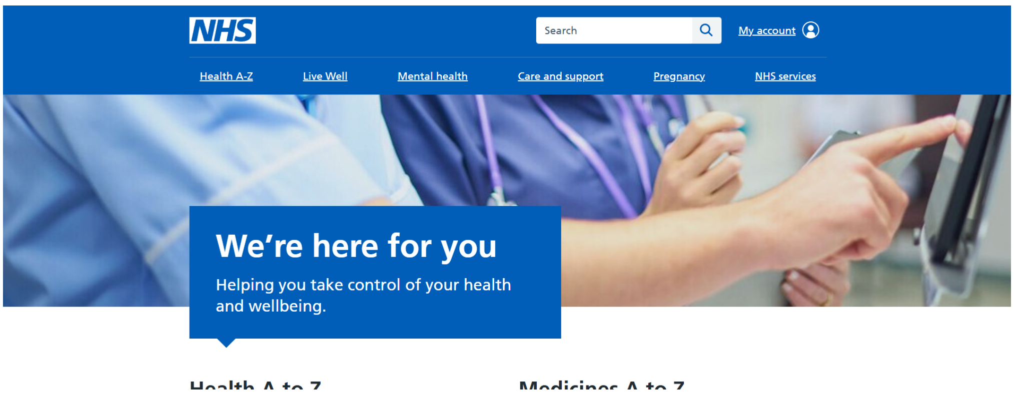

Figure 1.2: *NHS website layout from 2023.*

1.6 Structure of the Dissertation

The study will follow a structured, sequential approach to investigate the application of LLMs in personalised web content delivery. First, a literature review will be conducted to explore the foundations of user characteristics, personalisation strategies, web design principles, and the role of artificial intelligence—particularly LLMs and vision-language models—in adaptive systems.

Following the review, a prototype will be constructed that generates personalised health advice based on basic user characteristics such as age and gender. This prototype will integrate a LLM to produce text content and a vision-language model to match relevant images, automatically formatting the results into a complete webpage.

Once development is complete, the system will undergo a series of system-level experiments to evaluate its functional stability, content generation performance, and visual coherence. These tests will be followed by a user-based evaluation involving ten participants, who will compare the adaptive prototype with a static version and provide both quantitative ratings and qualitative feedback.

The collected data will then be analysed to assess the prototype's effectiveness in terms of user relevance, satisfaction, and usability. Finally, conclusions will be drawn based on the results, and recommendations will be made for further improvement and future research in adaptive content delivery using LLMs.

Chapter 2: Literature Review

To ground the study within existing research, this chapter critically reviews prior work on user personalisation, web design, and language model integration in health information systems.

The literature underpinning this study spans user diversity, personalisation technologies, and the integration of LLMs into adaptive systems. The review begins by examining user characteristics, highlighting how factors such as age, gender, cognitive preferences, and cultural background shape interactions with web content. This discussion also outlines key design principles—such as user-centred design and cultural sensitivity—that guide the development of inclusive digital experiences.

The next section explores the application of LLMs in personalising content to meet diverse user needs. It considers their role in improving engagement and relevance, while also addressing deployment challenges such as computational demands, data privacy, and prompt sensitivity.

Finally, the review draws on existing studies that combine personalisation, web design, and AI integration. By evaluating methodologies and outcomes across related work, it identifies current research gaps and positions this dissertation as a contribution to the evolving field of adaptive web content delivery.

2.1 User Characteristics

2.1.1 Introduction to User Characteristics in Web Design

Understanding user characteristics is fundamental to the design of effective and inclusive web interfaces. User characteristics, including demographic variables such as age, gender, and technological expertise, profoundly influence how individuals perceive, interact with, and experience web content [7]. A nuanced understanding of these factors can help designers create systems that cater to diverse user needs, improving usability, accessibility, and overall satisfaction.

Age and Technological Experience

Age is one of the most significant variables affecting user interaction with digital systems [7]. Older adults often face distinct challenges when interacting with web technologies, primarily due to differences in prior exposure and cognitive processing capabilities. Research highlights that older users tend to rely heavily on prior knowledge and external aids, such as manuals or clear instructions, during digital interactions. This reliance is linked to their limited exposure to evolving digital technologies and a higher cognitive load when adapting to new systems [7]. For instance, older adults often struggle with features requiring multitasking or rapid adaptation to novel interaction paradigms. However, their extensive life experience provides them with a broad repertoire of procedural and semantic knowledge that can be leveraged when designing interfaces [7].

Younger users, on the other hand, generally exhibit greater ease and adaptability with web systems [7]. Having grown up in an environment saturated with digital technologies, they often demonstrate higher confidence and efficiency in navigating web interfaces. Younger users are also more likely to experiment with new functionalities, showing a propensity for trial-and-error learning [7]. This demographic's familiarity with digital tools allows designers to incorporate more complex or innovative features without risking usability issues, as these users tend to adapt quickly.

However, technological experience, regardless of age, plays a critical role. Older adults with extensive technology experience, often referred to as ”silver surfers,” exhibit interaction patterns comparable to younger users. These findings underscore the importance of designing systems that accommodate not only age but also varying levels of technological proficiency [7].

Gender and Interaction Styles

Gender differences in web interaction are another critical consideration in web design [8]. Research consistently demonstrates that males and females approach web usage with distinct strategies and preferences. Males often adopt more exploratory and heuristic approaches, characterised by rapid navigation and a focus on achieving goals efficiently. They tend to engage in trial-and-error behaviour and are more likely to interact with multiple features simultaneously [8], [1]. This exploratory nature aligns well with designs that encourage flexibility and offer multiple pathways to access information. Figures 2.1, 2.2 and 2.3 further illustrate these interaction differences, specifically in the context of web search behaviours.

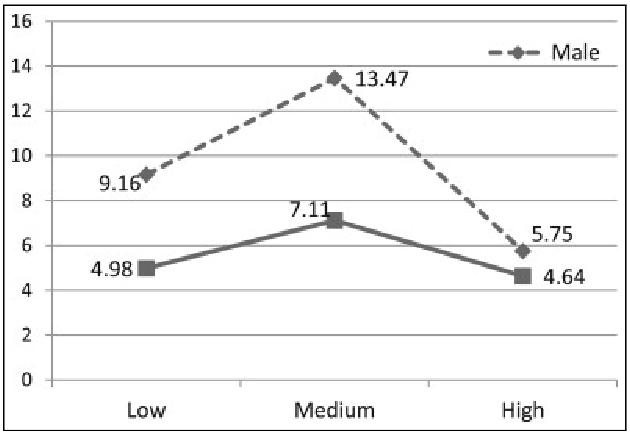

Figure 2.1: *Number of search queries conducted by males and females across low, medium, and high-performance groups, revealing that males conduct more search queries, particularly in the medium-performance category [1].*

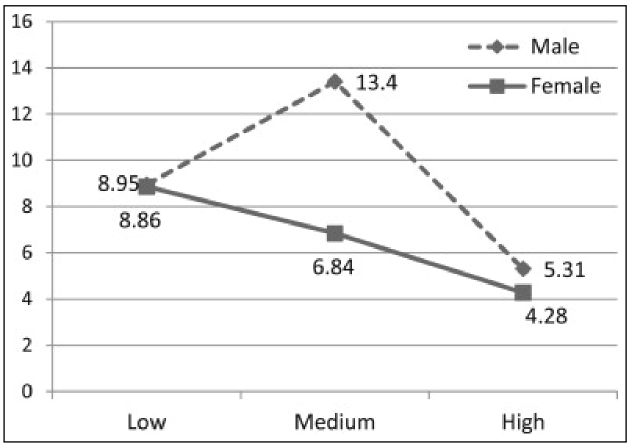

Figure 2.2: *Differences in search query updates, where males demonstrate greater engagement in modifying their queries, especially at medium performance levels [1].*

Conversely, females are generally more detail-oriented and deliberative in their

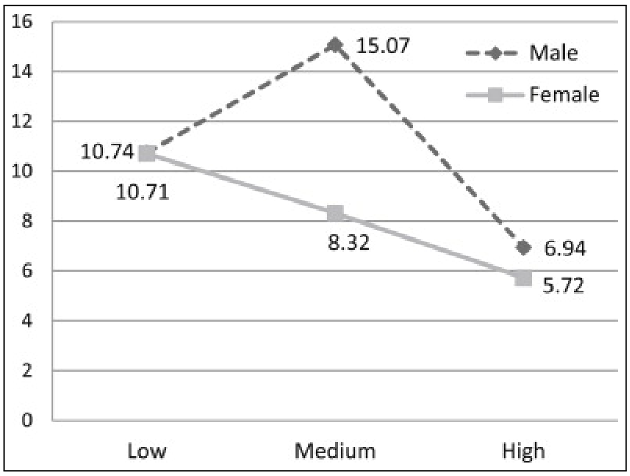

Figure 2.3: *Total number of searches conducted, further underscoring the exploratory tendencies of males [1].*

web interactions [8]. They often spend more time evaluating information and are likely to prefer interfaces that provide clear, structured, and logically organised content. Additionally, women tend to place a higher emphasis on visual and aesthetic elements, including colour schemes and layout consistency [8]. These preferences highlight the need for designs that balance functionality with visual appeal, ensuring that the interface meets the aesthetic and usability expectations of diverse user groups.

Interestingly, gender differences also extend to specific types of web use [1]. For instance, males are more likely to explore non-linear navigation paths, whereas females tend to follow linear and logical sequences when interacting with content. Such differences have implications for designing navigation menus, content organisation, and even search functionalities [1]. Incorporating flexible navigation systems that cater to both exploratory and structured approaches can significantly enhance user satisfaction.

Cognitive and Perceptual Factors

Beyond demographic characteristics, cognitive and perceptual factors also play a pivotal role in shaping user experience [9]. Factors such as visual preferences, perceived ease of use, and hedonic qualities such as enjoyment and aesthetic appeal, are central to user satisfaction and engagement [9]. Research has shown that users' perceptions of a website's usefulness and aesthetic appeal often influence their intention to continue using the site. For example, websites that balance functional elements with visual attractiveness are more likely to retain users and foster trust [9].

Perceived ease of use is another critical factor, particularly for users with limited technological expertise [9]. Simplifying interactions through intuitive design, clear instructions, and user-friendly interfaces can significantly reduce cognitive load and improve usability. Additionally, hedonic qualities, such as engaging visuals and enjoyable interactions, contribute to the overall user experience, encouraging repeated use and deeper engagement [9], [10].

Design Implications

The diversity in user characteristics necessitates a design approach that prioritises inclusivity and adaptability. Key strategies include customisable interfaces, intuitive navigation, and accessible content [7]. For example, providing options for users to personalise their interaction experience—such as adjustable font sizes, alternative colour schemes, or simplified layouts—can cater to a wide range of preferences and needs [7].

Designers must also consider how users' prior knowledge and interaction styles influence their behaviour. Leveraging familiar design patterns can ease the learning curve for novice users, while incorporating advanced features for experienced users can enhance engagement [7]. Similarly, balancing exploratory pathways with structured guidance ensures that the interface accommodates both heuristic and detail-oriented navigation styles [7], [9].

Furthermore, it is essential to address the unique challenges faced by specific user groups. For older adults, this might involve creating interfaces that support clear instructions, minimal cognitive effort, and reliance on prior knowledge [8]. For younger users, incorporating interactive and dynamic elements can enhance engagement. Gender differences can be addressed by designing interfaces that balance aesthetics with functionality, ensuring that both exploratory and structured preferences are met [8], [1].

2.1.2 Designing Web Content for Diverse User Characteristics

Creating effective web content for diverse audiences involves understanding and addressing variations in user characteristics, such as culture, cognitive preferences, and technological expertise [11]. Research emphasizes that visual design elements, particularly color schemes and overall aesthetics, play a crucial role in shaping user satisfaction and engagement. For instance, different colors evoke distinct emotional and psychological reactions, which can significantly influence user perceptions of usability and trustworthiness. Selecting culturally aligned color schemes and ensuring aesthetic coherence enhances emotional engagement and user retention [12].

Studies have shown that minimalist designs, characterized by limited color palettes and harmonious layouts, generally lead to higher user satisfaction due to their clarity and ease of navigation. In contrast, vibrant designs with bold, dynamic color schemes may evoke stronger emotional responses but can risk overwhelming users, particularly on information-heavy platforms. Balancing visual appeal with functional usability is essential for creating a satisfying user experience [12]. Furthermore, usability testing and participant feedback reveal that designs integrating culturally and psychologically resonant color schemes facilitate improved task efficiency and reduce cognitive load, fostering positive interactions with the website [12].

Understanding these dynamics is critical for designers aiming to create inclusive web experiences that cater to a broad range of users. By aligning visual elements with user characteristics and preferences, web designers can effectively enhance usability, satisfaction, and engagement across diverse audiences.

User-Centred Design Principles

User-centred design (UCD) prioritises tailoring websites to meet the needs of diverse user groups. Effective UCD principles focus on layout, navigation, and content structure to improve usability and satisfaction [11]. Studies indicate that intuitive navigation and structured layouts are critical for reducing user frustration and enhancing the overall experience. Personalisation further increases satisfaction by aligning content with individual user characteristics, improving engagement and task success rates [11].

Cultural Sensitivity in Design

Culture significantly influences user interactions with web content [13]. Individualistic cultures often favour features that support personalisation and autonomy, while collectivist cultures value designs that promote community and shared experiences. Additionally, the use of culturally relevant imagery and language enhances trust and engagement with websites. These cultural design considerations are especially important for global platforms aiming to serve diverse audiences effectively [13].

Leveraging Digital Nudging

Digital nudging employs subtle interface design elements to guide user decisions without restricting their freedom of choice [14]. Techniques such as emphasising default options, structuring content hierarchically, and using progressive disclosure help users navigate decision-making processes more efficiently. E-commerce platforms, for instance, leverage nudging by highlighting best-selling products or recommending actions to simplify decision-making [14]. These strategies improve user satisfaction while influencing behaviour in a predictable and user-beneficial manner.

Addressing Cognitive and Behavioural Variations

Cognitive styles and behavioural patterns also play a crucial role in web design. For instance, users with limited digital experience benefit from clear guidance and visual cues, while experienced users prefer flexible, open-ended interaction pathways [15]. Hypermedia systems and non-linear navigation structures accommodate these variations by allowing users to process information at their own pace and follow personalised routes. Effective design adapts to both novice and expert users by dynamically adjusting interface elements based on user behaviour [15].

Designing Web Content for Age-Specific Characteristics

Designing web content for age-specific characteristics requires careful consideration of typography to accommodate varying visual and cognitive needs across age groups. For older adults, larger font sizes enhance legibility and reduce reading strain, with 14-point fonts being particularly effective for improving readability and user experience [16]. Additionally, sans serif fonts are often preferred for their clarity and modern appearance, although serif fonts may facilitate faster reading in certain contexts [16]. By optimizing font type and size, web designers can create more accessible and inclusive interfaces that cater to the diverse needs of different age groups [16].

Enhancing the Web Experience

The quality of the web experience depends on integrating usability, functionality, and psychological design elements [17]. Successful websites combine interactivity, visual consistency, and responsive features to meet diverse user expectations [17]. Personalised recommendations, adaptive interfaces, and cross-platform compatibility are critical for engaging users and ensuring inclusivity. Additionally, feedback mechanisms enable designers to refine content iteratively based on real-world usage patterns [17].

2.2 Technology

2.2.1 Role of LLMs in Personalisation

LLMsn have emerged as transformative tools in artificial intelligence, demonstrating unparalleled capabilities in natural language processing, contextual understanding, and text generation [2]. Built on advanced transformer architectures, these models process vast datasets to deliver human-like outputs, making them essential for personalisation systems. As shown in Figure 2.4, the chronological progression of LLM releases highlights the rapid evolution of these models over recent years [2].

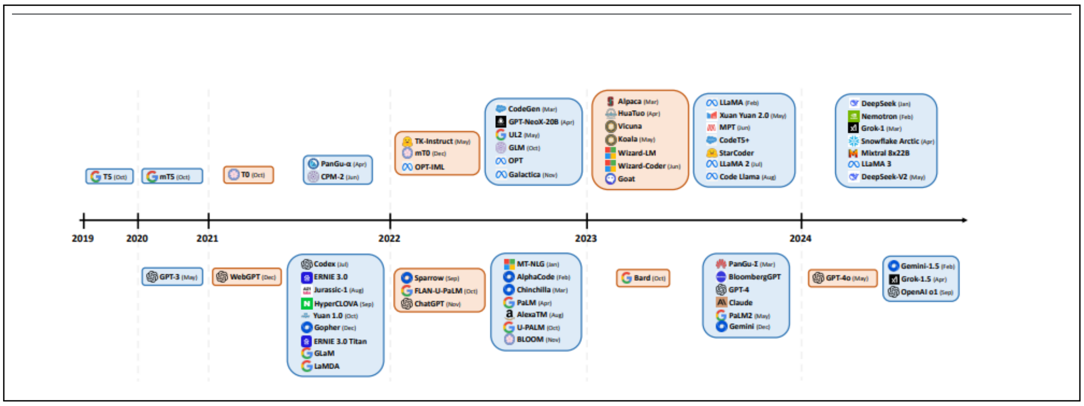

Figure 2.4: *The chronological display of LLM releases: blue cards represent 'pre-trained' models, while orange cards correspond to 'instruction-tuned' models. Models on the upper half signify open-source availability, whereas those on the bottom are closed-source. [2].*

Dynamic Personalisation through Content Generation

Traditional personalisation systems often rely on static algorithms, such as collaborative filtering or content-based filtering, to recommend products or structure user interactions [18]. While effective in certain contexts, these methods lack the flexibility to adapt to complex and evolving user behaviours. LLMs address these limitations by dynamically generating contextually relevant content [18]. For instance, in an e-commerce setting, LLMs analyse browsing history, purchase patterns, and search queries to deliver highly precise recommendations [18].

Unlike conventional systems, LLMs do not merely apply predefined rules; instead, they synthesise insights from diverse datasets to offer richer and more personalised experiences [2]. An example can be seen in their ability to generate tailored promotional messages, personalised reviews, and customised recommendations that align with individual user characteristics [2]. Such capabilities allow platforms to engage users more effectively and foster long-term loyalty.

Enhancing User Interaction

LLMs enable adaptive interactions that go beyond static communication. By interpreting conversational queries with remarkable accuracy, these models deliver responses that feel natural and context aware [19]. In customer service applications, for instance, LLM-powered chatbots can understand nuanced user concerns and provide tailored solutions in real time. This level of interaction improves user satisfaction and builds trust, key components of successful digital platforms [19].

Additionally, LLMs excel in dynamic question-answering systems, refining their responses based on iterative feedback [20]. A user searching for holiday destinations might initially receive general suggestions, but as preferences and constraints become clearer through interaction, the recommendations become more tailored [20]. This adaptive capacity mirrors the effectiveness of human assistants, creating a user experience that feels personalised and intuitive [20].

Personalised Recommendations and Multi-Modal Integration

LLMs are redefining the scope of recommendations by integrating multi-modal data streams, including text, images, and audio [21]. These models analyse user activity to provide tailored suggestions that extend beyond traditional formats. For example, in a streaming platform, LLMs might recommend content based on viewing history, time of day, or inferred user moods. By combining these insights with visual and auditory preferences, LLMs create a seamless and engaging user experience [21].

Table 2.1 compares these interaction methods based on user accessibility, input complexity, response accuracy, system requirements, and scalability [21]. The data demonstrates that while text-based interactions offer the highest accessibility and scalability, they are limited in response accuracy and require relatively low system resources. Conversely, immersive methods, such as VR/AR, provide high response accuracy but demand substantial system resources and have lower scalability [21].

Interaction Mode	User Accessibility	Input Complexity	Response Accuracy	System Requirements	Scalability
Text-based	High	Low	Moderate	Low	High
Voice-based	Moderate	Moderate	High	Moderate	High
Video-based	Low	High	High	High	Low
Immersive (VR/AR)	Low	High	High	Very High	Low

Table 2.1: *Comparison of Multi-modal LLM Interaction Methods*

In education, LLMs personalise learning paths by adapting content to individual students' proficiency levels and interests [22]. This is achieved through real-time analysis of student progress, enabling the generation of exercises, study guides, and feedback tailored to each learner [22]. Similarly, in healthcare, LLMs provide personalised wellness recommendations by synthesising patient histories and relevant medical literature, enhancing both accessibility and efficacy [23].

Transforming Web Interfaces with LLMs

Personalisation extends to web interfaces, where LLMs dynamically adjust content layouts and structures to match user characteristics [24]. For instance, homepages can be customised based on frequently accessed sections, and navigation menus can be restructured to prioritise user-specific needs. By doing so, LLMs create interfaces that are not only visually appealing but also functionally optimised for individual users [24].

In digital ecosystems, LLMs act as integrative tools, connecting various services to provide cohesive personalisation [25]. For instance, a healthcare portal might use LLMs to integrate data from multiple providers, presenting users with tailored recommendations such as medication reminders, fitness plans, or diet suggestions. Such integrations enhance user convenience while delivering a consistent and unified experience [25].

Expanding the Horizon of Personalisation

The potential of LLMs in personalisation continues to grow with advancements in fine-tuning and contextual adaptation [26]. Future developments may include integrating real-time environmental data, such as location or weather, to offer highly context-aware recommendations. Multi-modal models, which process text alongside visual and auditory data, hold the potential to create even richer and more immersive personalised experiences [26]. For instance, a retail platform could recommend outfits by combining user browsing behaviour with seasonal trends and visual preferences, delivering suggestions alongside styled images [26].

As these capabilities expand, LLMs are set to redefine user expectations, creating digital interactions that are not only more engaging but also more aligned with individual needs and preferences [26]. By advancing personalisation to unprecedented levels, LLMs pave the way for digital platforms to deliver experiences that are dynamic, adaptive, and deeply meaningful.

2.2.2 Challenges of LLMs in Personalisation

While LLMs have revolutionised personalisation in digital systems, their implementation is not without significant challenges [18]. Issues such as privacy concerns, computational inefficiencies, bias, and alignment with user expectations present barriers to their effective integration. These challenges must be addressed to fully realise the potential of LLMs in creating personalised user experiences.

Privacy and Data Security

Personalisation systems powered by LLMs require extensive data to deliver tailored recommendations and interactions [18]. This reliance on user data introduces critical privacy and security challenges. For instance, LLMs often handle sensitive user information, raising concerns about compliance with regulations like the General Data Protection Regulation (GDPR). Without robust safeguards, there is a risk of data leaks or misuse, which could erode user trust and lead to reputational and legal consequences [18], [27].

One specific risk is the inadvertent memorisation of sensitive information within the training process, a phenomenon that could lead to such data being revealed during model inference [27]. Addressing this requires implementing strategies such as data anonymisation and differential privacy, which can minimise the exposure of identifiable information while maintaining model performance [27].

Computational Overhead and Resource Demands

LLMs are resource-intensive, requiring significant computational power and memory for training and deployment [2]. These requirements pose a substantial barrier for smaller organisations or individual developers, who may lack the necessary infrastructure. Furthermore, deploying LLMs in real-time personalisation systems introduces latency issues, which can degrade user experience [2].

Efforts to mitigate these demands include techniques like model compression and pruning, which aim to reduce model size without significantly impacting performance [8]. However, achieving a balance between efficiency and accuracy remains an ongoing challenge. Researchers are also exploring more lightweight architectures to optimise the deployment of LLMs in environments with constrained resources [2], [8].

Bias and Fairness

Bias in LLMs arises from imbalances in the datasets used during training, reflecting societal prejudices or overrepresenting certain demographics [27]. When deployed in personalisation systems, these biases can manifest in recommendations or interactions that inadvertently marginalise or stereotype specific user groups. For example, an LLM may prioritise content that aligns with dominant cultural norms, thereby limiting the diversity of user experiences [27], [28].

Mitigating bias requires more representative training datasets, rigorous evaluation protocols, and advanced techniques for detecting and reducing bias during model fine-tuning. Additionally, improving transparency in how LLMs generate personalised outputs can enhance accountability and user trust [28].

Alignment with User Expectations

Aligning LLM outputs with user expectations presents another critical challenge. Personalisation inherently involves adapting to dynamic user characteristics, which can evolve over time [29]. However, existing LLMs often struggle to capture these changes, leading to a mismatch between user needs and the recommendations or responses provided [29].

This challenge is compounded by the reliance on prompt engineering, which can introduce variability in model performance. Fine-tuning LLMs to align with user-specific contexts is resource-intensive and may lead to overfitting or reduced generalisability across diverse use cases [28]. Developing more robust evaluation frameworks is essential to ensure that personalisation systems consistently meet user expectations.

Evaluation and Scalability

Effective evaluation of LLMs in personalisation systems remains an underexplored area. Current benchmarks are often insufficient for capturing the nuances of personalised interactions, particularly in scenarios involving diverse and dynamic user bases [2]. Moreover, as LLMs grow in complexity, establishing standardised evaluation metrics that balance performance, fairness, and ethical considerations becomes increasingly critical [2].

Scalability also poses challenges, particularly for platforms serving large numbers of users with diverse preferences. Ensuring that personalised experiences remain consistent and equitable across such a wide spectrum requires innovations in model training, deployment, and adaptation [18], [29].

2.3 Similar Studies

The field of user-centred web design and personalisation has been extensively explored, offering insights into usability, user characteristics, and emotional engagement. Using the themes identified above, relevant studies were reviewed in relation to the themes of usability and structural design, emotional and psychological dimensions, cultural and contextual factors, and integration of flow and user-centred design.

2.3.1 Usability and Structural Design

Dianat et al. [11] investigated the relationship between web design attributes and user satisfaction within the context of Iranian online banking platforms. Using a multivariate regression analysis, they studied 798 participants who had used online banking for over a year. Key variables included web design factors like layout, navigation, and performance, as well as user demographics such as age and gender. Results indicated that web structure and layout were the most significant predictors of usability and satisfaction, while personal characteristics had minimal impact, underscoring the importance of optimised structural design for enhancing usability [11].

Lee and Koubek [30] analysed usability and design attributes in e-commerce websites using a mixed-methods approach. Their methodology combined user interviews with task-based usability testing to assess the influence of navigation, aesthetics, and functionality on user preference. Findings revealed that task-oriented layouts and structured navigation were more critical than aesthetic considerations. Together, these studies demonstrate the value of foundational structural elements in designing user-friendly and functional websites [30].

2.3.2 Emotional and Psychological Dimensions

Partala and Kallinen [3] explored user experiences through emotions and psychological needs using the Positive and Negative Affect Schedule (PANAS) and a customised psychological needs questionnaire. The study involved 45 participants describing satisfying and unsatisfying user interactions. Quantitative analysis revealed that positive experiences were associated with feelings of autonomy and competence, while negative experiences lacked these drivers. This highlights the importance of emotional and psychological engagement in user-centred design [3].

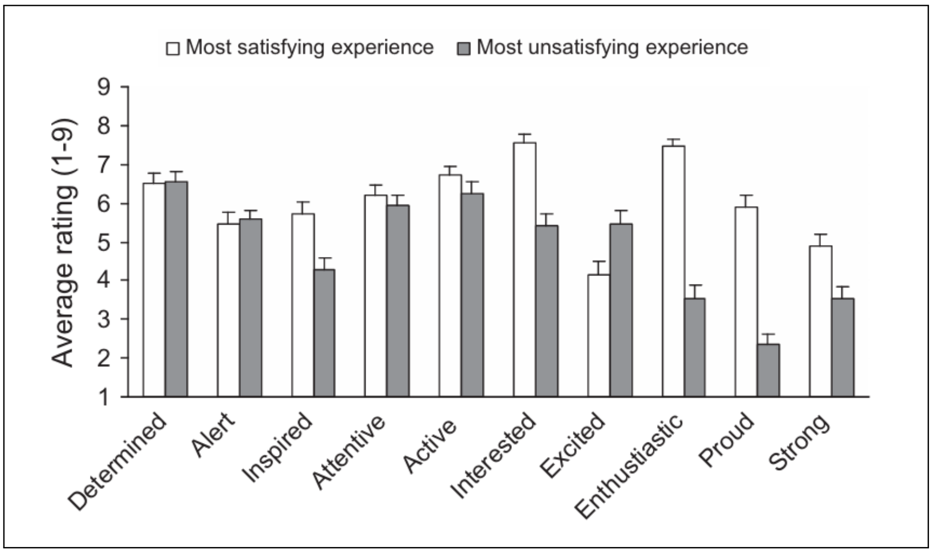

Figure 2.5: *Comparison of positive emotional responses reported during the most satisfying and unsatisfying user experiences, based on PANAS ratings [3].*

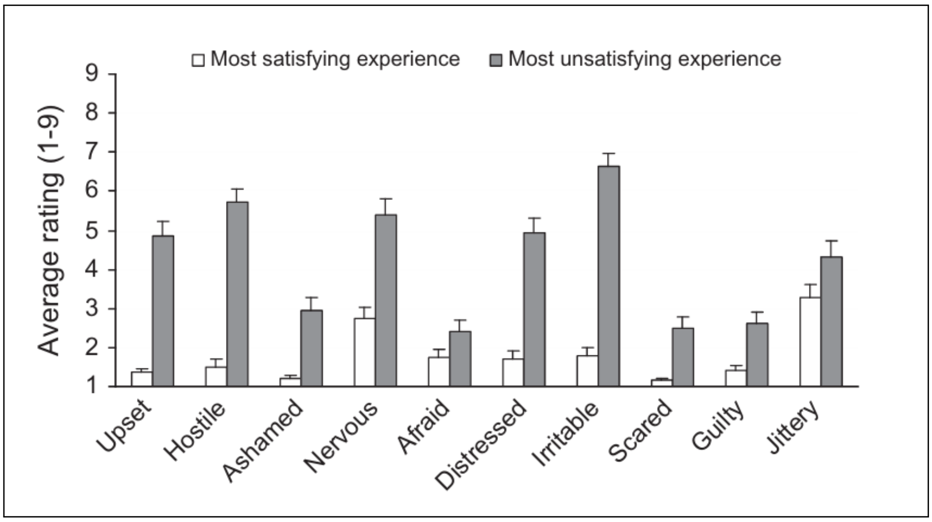

Figure 2.6: *Comparison of negative emotional responses during the most satisfying and unsatisfying user experiences [3].*

Skadberg and Kimmel [4] introduced the concept of ”flow experience” in web interactions. Their study employed observational techniques and self-reports to identify factors contributing to flow, such as interactivity, content relevance, and aesthetic coherence. Results demonstrated that immersive experiences increased user engagement and interaction duration. This reinforces the significance of interactivity and relevant content in fostering user satisfaction and loyalty [4].

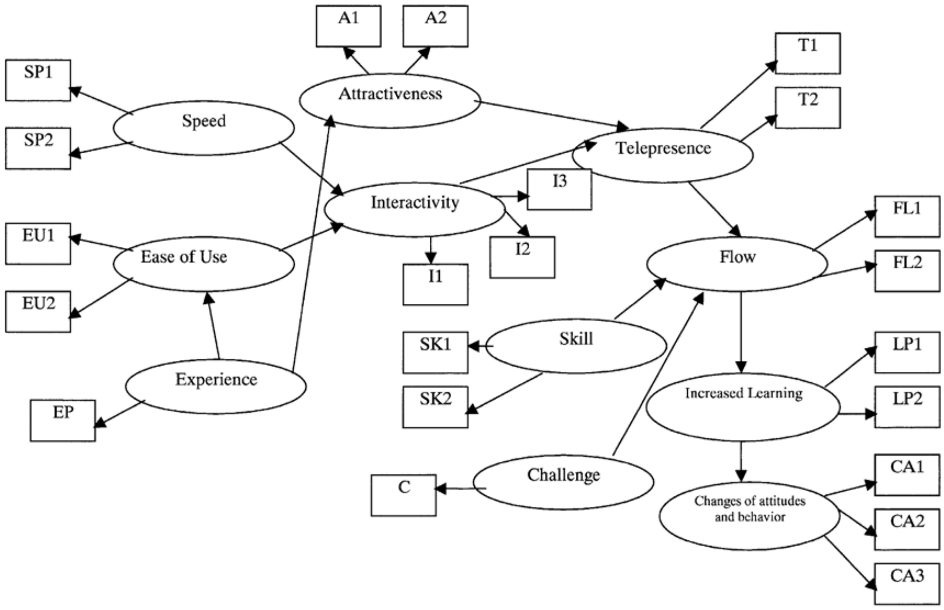

Figure 2.7: Skadberg and Kimmel's hypothesised flow model, showing how interactivity and usability factors influence flow experience, learning, and behavioural change [4].

In addition to emotional engagement, cultural and contextual differences further influence user interaction with digital content.

2.3.3 Cultural and Contextual Factors

Cultural and demographic variations also play a crucial role in shaping web design preferences. Dianat et al. [11] observed that while structural elements like navigation and layout are universally important, localisation to align with cultural norms significantly enhances user trust and satisfaction. This finding emphasises the need to combine universal usability principles with culturally sensitive adaptations to create inclusive digital platforms [11].

In a complementary perspective, Vu et al. [31] stressed the importance of tailoring web design to meet diverse cognitive and physical capabilities. Their study argued that understanding the goals of specific user demographics allows for the development of more accessible and engaging interfaces. These insights align with the broader advocacy for user-centred design approaches that prioritise inclusivity [31].

2.3.4 Integration of Flow and User-Centred Design

Skadberg and Kimmel's flow model integrates well with user-centred design principles, illustrating how usability, interactivity, and content relevance foster immersive user experiences. Their findings align with Partala and Kallinen's emphasis on emotional engagement, showing that combining functional and experiential elements leads to optimal user satisfaction. Together, these studies advocate for designs that balance utility and engagement to meet diverse user needs [3], [4].

2.3.5 Observations and Synthesis

The reviewed studies collectively underscore the importance of usability, emotional engagement, and cultural adaptability in web design. Dianat et al. [11] and Lee and Koubek [30] focused on structural and task-oriented design, while Partala and Kallinen [3] and Skadberg and Kimmel [4] explored experiential and emotional dimensions. Vu et al. [31] added insights into cultural and demographic considerations. These studies collectively provide a robust framework for understanding user-centred design and personalisation.

2.4 Concluding Summary

The literature reveals a strong link between user characteristics and web usability. Factors such as age, gender, cognitive style, and cultural background shape how users interact with web interfaces and influence their satisfaction with digital content. These findings underscore the importance of inclusive and adaptive design approaches, particularly those rooted in user-centred and culturally sensitive methodologies.

Advancements in artificial intelligence, particularly the emergence of LLMs present new opportunities to deliver personalised content that adapts to individual user needs. However, challenges such as computational demands, data privacy, and semantic relevance persist, especially when deploying these technologies in real-time web environments.

Existing studies show that adaptive systems can enhance perceived relevance, engagement, and trust in web-based health communication. Yet gaps remain in how effectively these systems are integrated with vision-language models, and in how they are evaluated across diverse user populations. These gaps inform the direction and methodology of the current study, which aims to investigate whether an LLM-powered prototype can meaningfully enhance the user experience compared to a static counterpart.

The reviewed studies highlight both the potential and challenges of personalised digital content. These insights directly informed the design decisions and methodological choices described in the following chapter.

Chapter 3: Research Methodology

3.1 Objectives

This study aims to develop an adaptive web content delivery system utilising LLMs to personalise user experiences. The study attempts to explore how user characteristics, such as age and gender, can be leveraged to tailor web content dynamically for enhanced usability and engagement.

Based on the stated aim of the study, the following objectives were established. Presented in a sequential order, these objectives collectively contribute to the implementation and evaluation of the proposed prototype.

- Develop a web-based prototype that delivers personalised health content using user characteristics such as age and gender.
- Integrate an LLM for generating tailored text content and a vision-language model for selecting semantically appropriate images.
- Evaluate the technical stability and consistency of the prototype through system-level tests.
- Compare the usability and perceived value of the personalised prototype against a static version via user testing.

- Analyse participant feedback and usage data to determine the impact of personalisation on user experience, relevance, and satisfaction.
- Identify limitations and propose improvements for future adaptive content delivery systems.

3.2 Research Philosophy and Design

This study adopts a pragmatic research philosophy, which is particularly well-suited to investigations that combine technical development with human-centred evaluation. Pragmatism allows for the integration of both objective and subjective data, focusing on the practical outcomes of research rather than adherence to a single philosophical paradigm [32], [33]. Given the nature of this study—developing and evaluating an adaptive web content delivery system using LLMs to personalise content based on user characteristics—a pragmatic stance enables a balanced consideration of both system performance and user experience.

Correspondingly, a mixed-methods research design is employed. This approach combines elements of both quantitative and qualitative methodologies to achieve a more comprehensive understanding of the system's effectiveness. Mixed-methods designs are particularly valuable when evaluating interactive systems, as they accommodate both empirical performance metrics and human-centred feedback [34], [35]. Quantitative methods are used to measure system performance and user interaction metrics such as content relevance ratings, engagement time, or response accuracy, while qualitative methods are applied to gather user feedback and explore individual perceptions of content personalisation. This dual approach ensures that the research not only evaluates the technical efficacy of the LLM-generated system but also captures the nuanced experiences of its users.

3.3 Pipeline Diagram

The pipeline for this prototype is divided into three main phases, covering the entire process from user interaction to evaluation. It begins with the user submitting their dynamic characteristics through a Gradio interface, after which the inputs are validated and used to retrieve personalised health tips from the MyHealthfinder API.

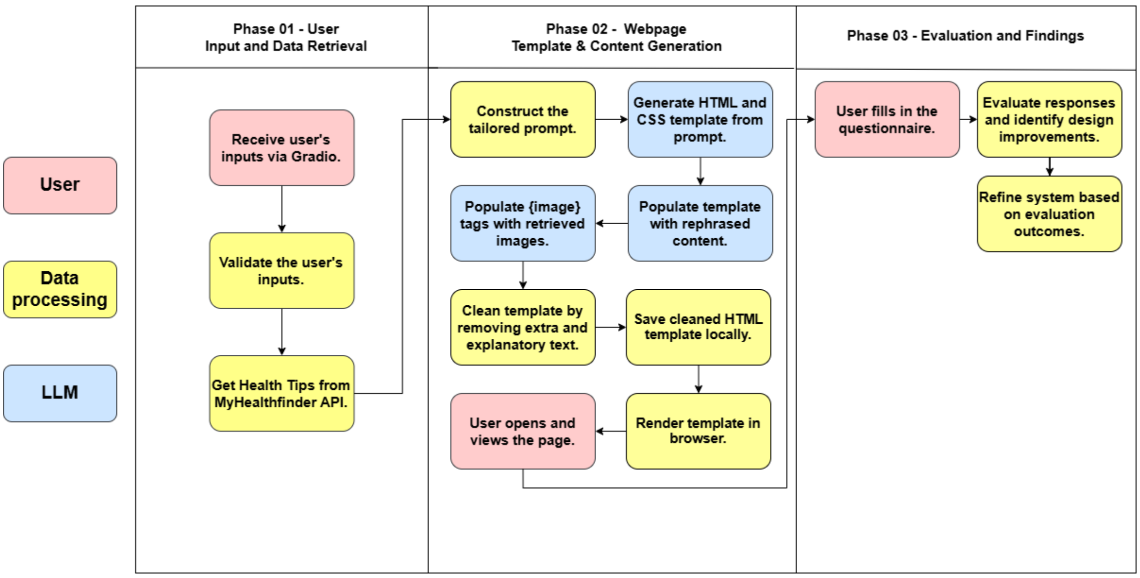

Figure 3.1: *Pipeline Diagram.*

In the second phase, a tailored prompt is constructed and passed to a LLM, which generates the initial HTML structure. This template is then populated with relevant images and rephrased health content before being cleaned and styled using Bootstrap. The finalised HTML page is saved locally and can be opened in the browser for the user to view.

Finally, in the evaluation phase, the user completes a questionnaire based on their experience, and the responses are collected for analysis.

3.4 Prototype Core Algorithms

3.4.1 User Input Handling and API URL Construction

This algorithm initiates the personalisation process by collecting demographic data from the user through a Gradio interface. The user is prompted to enter their characteristics, which are then used to dynamically construct a request URL targeting the MyHealthFinder public API. The structure of the URL incorporates the user inputs as query parameters, allowing the system to retrieve a personalised set of health recommendations in real time. The Gradio interface can be seen in Figure 3.2 below.

Once the API response is received in JSON format, the algorithm extracts relevant fields including the health tip titles and their corresponding descriptions. Since the API often includes HTML formatting tags within descriptions, the algorithm uses regular expressions to clean the text and ensure the output is plain and readable. This results in a structured list of health tips ready for downstream pro-

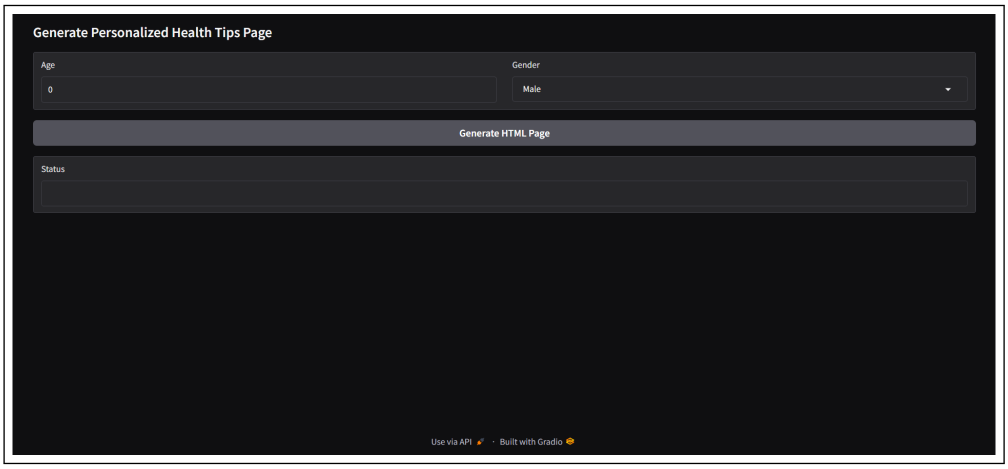

Figure 3.2: *The Gradio interface where users input their characteristics.*

cessing. The API retrieval and data-cleaning functionality is encapsulated within a single function, which serves as the entry point for generating personalised content. Below is a code snippet of this function.

```
def get_health_tips(age, gender):
    gender = gender.lower()
    url = f"https://odphp.health.gov/myhealthfinder
    /api/v3/myhealthfinder.json?age={age}&sex={gender}"

    try:
        response = requests.get(url)
        data = response.json()
        resources = data["Result"]["Resources"]["all"]["Resource"]
        tips = []
        for res in resources:
            title = res.get("MyHFTitle", "No Title")
            desc_html = res.get("MyHFDescription", "")
            desc = re.sub(r"<.*?>", "", desc_html).strip()
            tips.append((title, desc))
        return random.sample(tips, k=min(5, len(tips)))
    except Exception as e:
        print(f"API error: {e}")
```

```
        return []
```

3.4.2 Image Captioning with BLIP

The second algorithm focuses on generating descriptive captions for locally stored images using the BLIP (Bootstrapped Language-Image Pretraining) model. This step is necessary because the images themselves lack metadata or alternative text that could be used for matching. By applying a pretrained vision-language model, the system is able to generate meaningful, context-aware textual descriptions for each image.

```
def generate_image_captions():
    captions = {}
    for filename in os.listdir(IMAGE_DIR):
        if filename.lower().endswith((".jpg", ".jpeg", ".png")):
            path = os.path.join(IMAGE_DIR, filename)
            try:
                image = Image.open(path).convert("RGB")
                inputs = blip_processor(image,
                    return_tensors="pt").to(blip_model.device)
                out = blip_model.generate(**inputs)
                caption = blip_processor.decode(out[0], skip_special_tokens=True)
                captions[path] = caption
            except Exception as e:
                print(f"Skipping {filename}: {e}")
    return captions
```

The implementation iterates through a directory of image files, loads each one, and processes it through the BLIP captioning pipeline. The resulting caption is decoded and stored in a dictionary where the file path is mapped to the generated caption. These captions provide semantic content that can later be used to establish relationships between visual assets and the health tips retrieved earlier. This process removes the need for manual labeling and enables scalable visual personalisation.

3.4.3 Image and Health Tip Matching

After captions have been generated for each image and the list of health tips has been finalised, this algorithm matches each tip with the most semantically appropriate image. The matching process uses a sentence embedding model, all-MiniLM-L6-v2, from the SentenceTransformer library to convert both image captions and health tips into high-dimensional vectors within the same embedding space. The code below outlines this logic.

```
def match_tips_with_images(tips, image_captions):
    used = set()
    tip_texts = [f"{t[0]}. {t[1]}" for t in tips]
    tip_embeddings = embedder.encode(tip_texts, convert_to_tensor=True)

    image_texts = list(image_captions.values())
    image_paths = list(image_captions.keys())
    image_embeddings = embedder.encode(image_texts, convert_to_tensor=True)

    results = []

    for i, tip_emb in enumerate(tip_embeddings):
        sims = util.cos_sim(tip_emb, image_embeddings)[0]
        sorted_indices = torch.argsort(sims, descending=True)
```

```
        for idx in sorted_indices:
            img_path = image_paths[idx]
            if img_path not in used:
                used.add(img_path)
                results.append({
                    "title": tips[i][0],
                    "description": tips[i][1],
                    "image": img_path.replace("\\", "/")
                })
                break
    return results
```

Once embedded, the cosine similarity between each health tip and all available image captions is computed. The image with the highest similarity score that has not yet been used is selected for pairing. This ensures that each image is assigned uniquely, avoiding repetition and enhancing diversity. By performing this semantic comparison, the system is able to present each user with content that is not only textually relevant but also visually aligned.

3.4.4 HTML Post-Processing and Cleaning

The final core algorithm is responsible for refining the HTML code generated by the DeepSeek language model. The raw output from the model typically includes valid content, but it may also contain extraneous text, improperly formatted tags, or unused elements such as JavaScript blocks. This algorithm addresses these issues through several stages of post-processing.

It begins by extracting valid HTML between the document's opening and closing tags, followed by the removal of any embedded JavaScript. Image elements are reformatted with inline styles and Bootstrap classes to ensure consistent sizing and responsiveness. A MyHealthFinder badge is also appended near the bottom of the page as a visual anchor and citation. In addition, the algorithm integrates Bootstrap by injecting the necessary CDN links into the header and restructuring the body for improved layout. Depending on the user's gender input, a color scheme is applied to enhance personalisation. Finally, each health tip block is wrapped in a Bootstrap card component, improving readability and visual coherence.

3.5 Technologies Used

3.5.1 Hardware Specifications used for Testing and Analysis

Component	**Specification**
Processor	AMD Ryzen 7 5800H with Radeon Graphics, 3201 MHz, 8 Core(s), 16 Logical Processor(s)
RAM	Available 16GB
Graphics	AMD Radeon(TM) Graphics (No CUDA Support)

Table 3.1: *Hardware Specifications used for Testing and Analysis*

3.5.2 Software Specifications

Software	**Description**
Python 3	The main programming language used to implement the entire backend logic and data processing pipeline.
Ollama	A local LLM serving tool used to run the deepseek-r1:7b language model for generating HTML content.
Gradio	Used to build the web-based interface for collecting user inputs and displaying system feedback.
Visual Studio Code	Used for code development and debugging.

Table 3.2: *Software Specifications*

3.5.3 Python Libraries and Models

Library/Model	**Usage**
Gradio	For building the user interface (gradio==3.x).
Requests	For sending HTTP requests to the MyHealthFinder API.
re (Regex)	For cleaning HTML tags from the API's text descriptions.
Torch	Used as the backend for deep learning models like BLIP.
BLIPProcessor and BLIPForConditional-Generation	From Salesforce/blip-image-captioning-base: For generating captions from images.
Pillow (PIL)	For image loading and preprocessing before passing to BLIP.
all-MiniLM-L6-v2	For generating vector embeddings of health tips and image captions.
Subprocess	For calling the Ollama CLI and executing LLM queries.
Random	For sampling a subset of health tips when more are returned than needed.
JSON	For parsing API responses.
OS	For managing file directories and paths.

Table 3.3: *Python Libraries and Models Used*

3.6 Experiment Design and Sampling Strategy

This study adopts a mixed-methods experimental design to evaluate the effectiveness of an adaptive web content delivery system powered by a LLM, in comparison to a traditional static alternative. The objective is to assess whether the LLM-enhanced prototype delivers more personalised and user-relevant health information based on user characteristics such as age and gender.

A convenience sampling strategy is employed to recruit participants, consisting of individuals from the researcher's immediate network, primarily family members. Although not randomised, the sample is intentionally selected to include participants of diverse ages and genders, which is central to evaluating the personalisation capabilities of the system.

The experimental procedure involves the following steps:

1. Participants begin by accessing a Google Form, where they provide their dynamic characteristics and are given instructions to visit the static prototype.

2. After viewing the static webpage, participants return to the questionnaire and answer a set of questions based on their experience with that version.

3. Next, participants are directed to view the LLM-generated prototype. This version is created dynamically based on their inputs and includes personalised health tips and visuals.

4. After interacting with the personalised prototype, participants return to the questionnaire to complete a second set of questions specifically about the LLM-generated version.

5. A final section of the form invites participants to reflect on and compare both versions, providing feedback on usability, personalisation, visual appeal, and overall preference.

Collected quantitative data will be analysed using descriptive statistics, while qualitative responses will be reviewed using basic thematic analysis to identify recurring feedback patterns regarding usability, personalisation, and content relevance.

This structure enables a focused evaluation of the LLM's impact on web content personalisation and provides insights into how adaptive delivery affects the user experience compared to a static alternative.

3.7 Ethical Consideration

This study adheres to established ethical guidelines for research involving participants. All participants were provided with detailed information regarding the study's purpose, procedures, and their rights, including the right to withdraw at any time without penalty. Informed consent was obtained in writing before participation.

To ensure privacy and data protection, all participant data was anonymised and securely stored in compliance with applicable data protection regulations, such as the GDPR. No personally identifiable information was collected or retained beyond what was necessary for the purposes of analysis.

Given the study's focus on adaptive content delivery based on user characteristics, additional care was taken to ensure that no sensitive or discriminatory profiling occurred. User characteristics were handled in a way that prioritised ethical design and inclusivity, aligning with best practices in HCI and AI research.

Chapter 4: Analysis of Results and Discussion

This chapter presents the outcomes of both system-level tests and user evaluations. The section begins with technical test runs to validate stability, followed by participant feedback gathered through structured usability testing.

4.1 System-Level Experiments

This section evaluates the technical stability and consistency of the adaptive content delivery system. By running the prototype multiple times with fixed inputs, system behaviour, generation time, and recurring issues were assessed across both the language and vision-language model components.

4.1.1 Prototype Stability Tests

To assess the stability and consistency of the adaptive web content delivery system, the prototype was tested across multiple dimensions, including generation success rate and layout consistency. The backend pipeline consisted of two core components:

- BLIP Processor: A pretrained vision-language model responsible for generating captions for a local image set and semantically matching them to health tips.

- DeepSeek r1:7b (via Ollama): The language model used to construct the HTML structure and apply stylistic and structural formatting based on personalised content.

The prototype demonstrated high reliability in terms of structural generation. In all tests conducted, the DeepSeek r1:7b model successfully produced readable, visually cohesive HTML documents without crashes or complete failures. The layout structure remained consistent, with all five health tips displayed, and no null or empty outputs were observed.

Visual issues were minimal but notable. Only one minor formatting anomaly was observed, and no visual elements caused rendering failures. However, some content variations, such as additional labels like “Title:” and “Description:”, suggest opportunities for refinement in prompt engineering or post-processing.

The most persistent reliability concern was tied to the BLIP image matching stage. Tip #5 was repeatedly associated with weak image alignment or failure to render an image at all. This pattern indicates a potential edge-case limitation in the semantic matching process, possibly due to a lack of diversity or clarity in the available image captions.

Overall, the system demonstrated structural and operational stability but revealed limitations in the robustness of its vision-language integration, particularly

Run	Time Taken	Content Issues	Visual Issues	Notes/Anomalies
1	133.4s	None	Minor	
2	116.0s	None	None	
3	113.3s	Minor	None	Tip #5 matched image does not match as much
4	127.0s	Minor	None	Tips have ”Title:” And ”Description:” showing
5	121.6s	None	None	
6	125.2s	Major	None	Tip #5 image does not exist
7	127.1s	None	None	
8	130.8s	Minor	None	Tip #5 matched image does not match as much
9	132.1s	None	None	
10	130.1s	None	None	

Table 4.1: *Summary of Observed Outcomes*

for consistently generating well-matched images for all tips.

4.1.2 Repetition Tests

To further examine the prototype's consistency, a series of ten repetition tests were conducted using identical user input parameters: age 20 and gender male. These controlled conditions allowed for observation of any non-deterministic behaviour or generation anomalies across identical requests.

Each run involved the generation of a complete HTML health advice page. Response times varied between 113.3 and 133.4 seconds, with an average of approximately 126 seconds. While acceptable in a development context, this latency may require optimisation for real-time or production use.

Figures 4.1, 4.2, and 4.3 present example outputs from three of the ten repetition tests. These screenshots show consistent layout generation across runs and minor differences in phrasing or image matching.

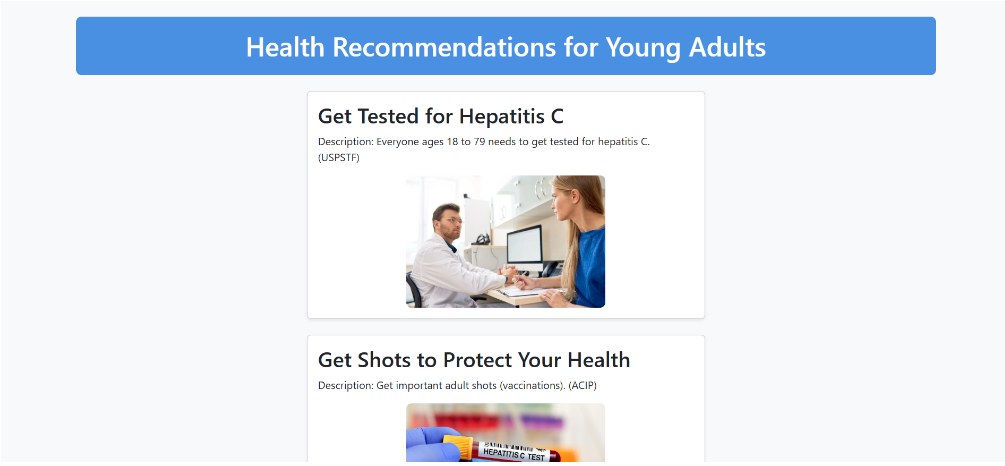

Figure 4.1: *LLM-Generated Prototype – Test Run 1 Output (Age: 20, Gender: Male)*

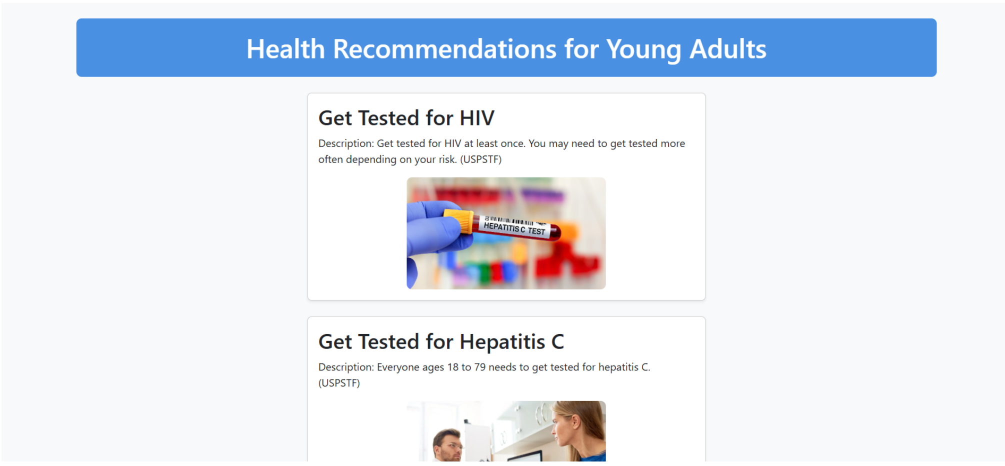

Figure 4.2: *LLM-Generated Prototype – Test Run 2 Output (Age: 20, Gender: Male)*

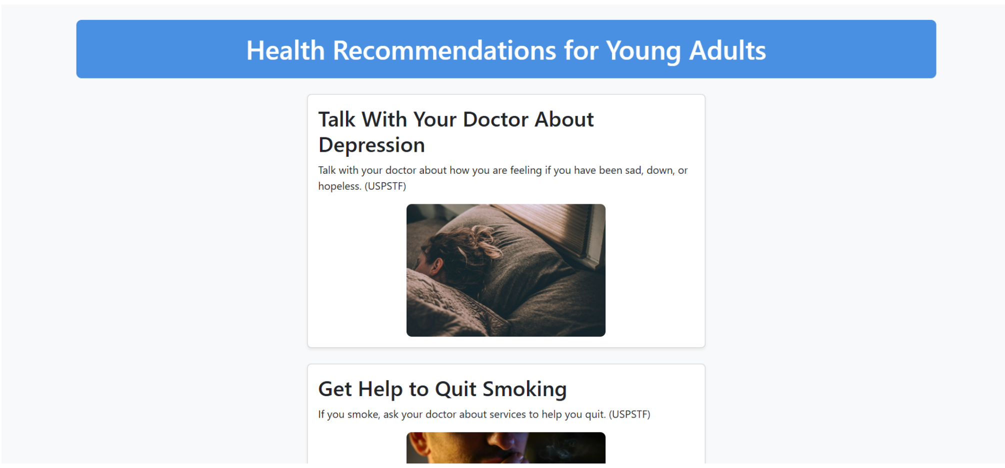

Figure 4.3: *LLM-Generated Prototype – Test Run 3 Output (Age: 20, Gender: Male)*

In Run 6, a complete image failure occurred for Tip #5, shown in Figure 4.4. While the rest of the page was generated successfully, this instance underscores the need for fallback handling or more precise image selection logic.

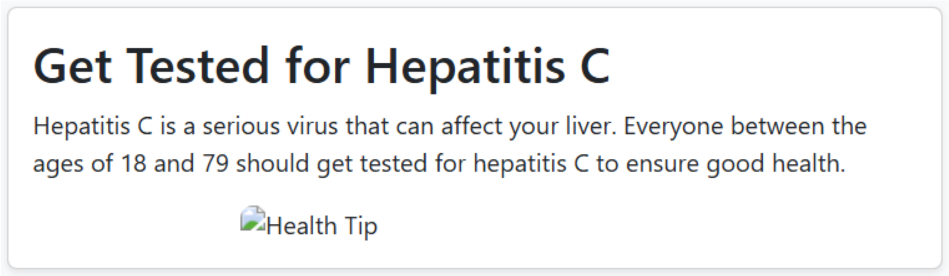

Figure 4.4: *Example from Run 6 showing Tip #5 with a missing image due to matching failure.*

Across all ten runs, the content remained functional and readable. However, minor non-deterministic variations in phrasing and image alignment were observed, primarily impacting Tip #5. These results confirm the system's ability to consistently generate structurally sound output, while highlighting opportunities to increase output determinism and visual accuracy through image pool expansion or stricter captioning criteria.

4.1.3 LLM Model Comparison

To explore the adaptability and robustness of the content generation pipeline, a comparative analysis was conducted between the default LLM model, DeepSeek r1:7b, and an alternative LLM—Gemma 3. The objective was to assess whether Gemma 3 could maintain or improve the quality of the generated healthcare webpages when tasked with rendering structured HTML from matched tip-image data.

Across four controlled runs using different user characteristics, Gemma 3 successfully generated semantically coherent and well-structured webpages in most cases. It demonstrated the ability to format HTML outputs with appropriate use of Bootstrap elements, gender-based theming, and responsive layout principles. However, the model also showed variability in content generation stability, with one instance resulting in a complete content failure, and another producing an output unexpectedly localised in Spanish. This suggests that while Gemma 3 is capable, it is more sensitive to prompt variation or internal defaults and requires tighter instruction handling.

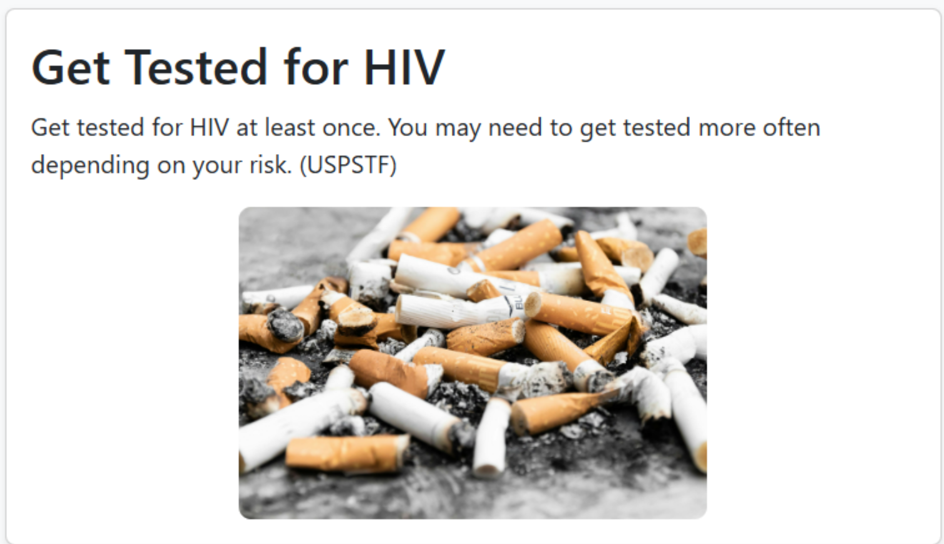

Figure 4.5: *Example of image–tip mismatch using Gemma 3: a smoking-related image is paired with an unrelated HIV test tip.*

In contrast, DeepSeek r1:7b demonstrated greater reliability in all tested scenarios. Across all ten prototype runs, it consistently produced HTML documents with the correct number of tips, structurally valid formatting, and gender-appropriate visual styling. Minor issues were mostly limited to occasional label formatting errors or weak image connections, but the model never failed to deliver a complete page.

Model Name	Run	Age	Gender	Tips Diverse?	Image Matches Accurate?	Visual Quality
Gemma 3	1	20	Male	Tips are varied. Well-suited to the target demographic.	Mostly. While most are relevant, two used generic or slightly mismatched images.	Good. Clean layout, consistent card format and responsive design.
Gemma 3	2	16	Female	Yes. The tips span physical exams, vaccines, and relationships, showing good range.	Partially accurate. Two mismatched images — others were mostly fine.	Strong. Layout uses Bootstrap well, and displays good spacing.
Gemma 3	3	59	Male	Yes. Covers cancer screenings, blood pressure etc. Appropriate and varied set for the age group.	Partially accurate. Two mismatched images — others were mostly fine.	Clean Bootstrap layout, properly structured, readable.
Gemma 3	4	40	Female	Yes, the tips include hepatitis C testing, drug misuse — good spread of physical, mental and nutritional health.	Some matches are appropriate. One image is weakly connected.	Layout is consistent and readable, though some sections could be better styled with Bootstrap.

Table 4.2: *Evaluation of Gemma 3 as replacement for DeepSeek r1:7b*

In conclusion, while Gemma 3 produced visually strong and semantically appropriate outputs in several cases, its occasional instability such as language switching and blank output makes it less dependable than DeepSeek r1:7b for consistent prototype delivery. DeepSeek r1:7b remains the more suitable LLM for this system, offering higher reliability and prompt adherence, which are critical for delivering personalised health information at scale.

Figure 4.6: *Correctly matched health tip and image: The caption "Get Help to Quit Smoking" is accurately paired with a contextual photo of a person smoking, reinforcing semantic relevance.*

4.1.4 Vision-Language Model Comparison

In addition to evaluating the impact of different LLMs, the prototype system was also tested with an alternative image captioning and matching model: OpenAI's CLIP. This was compared against the originally used model - BLIP (Bootstrapped Language Image Pretraining) - which was responsible for generating image captions and matching them to relevant health tips retrieved from the MyHealthFinder API.

The substitution aimed to determine whether CLIP could improve the semantic quality of image–text alignment, tip diversity, or layout consistency. Both models followed the same embedding-matching logic via SentenceTransformer and cosine similarity, ensuring that evaluation differences stemmed directly from captioning/-model output rather than downstream processing.

Across four runs using OpenAI CLIP, the results were mostly positive but revealed some consistency concerns. In three of the four runs, the tip diversity was judged to be moderate to good, with a reasonable distribution of physical, mental, and preventative health topics. However, one run failed to generate any health tips, resulting in a blank output. In contrast, BLIP consistently generated valid, diverse outputs in all ten baseline tests.

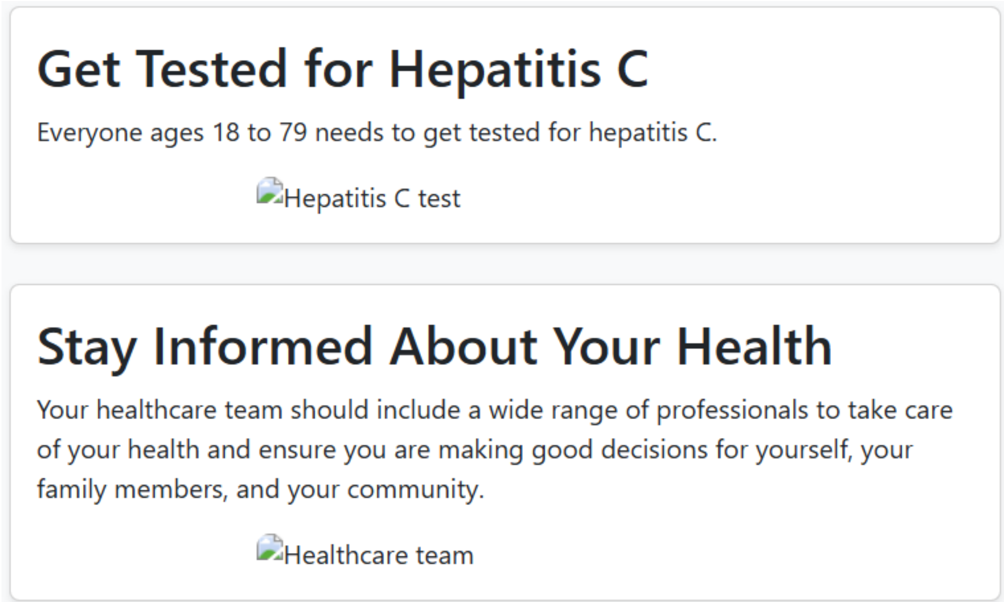

Figure 4.7: *Failed output in OpenAI CLIP run: no health tips were rendered despite successful page generation.*

Image–text alignment quality under CLIP was also slightly lower. While CLIP captions allowed for 3 to 4 accurate matches per run, it occasionally misrepresented health tips visually—e.g., matching a “Get tested for HIV” tip with an image labeled “HEPATITIS C TEST.” In contrast, BLIP generated captions that more precisely described visible content, likely due to its image-conditioned decoding process.

Model Name	Run	Inputted Age	Inputted Gender	Tips Diverse?	Image Matches Accurate?	Visual Quality
OpenAI CLIP	1	20	Male	Moderate: 3 of 5 tips were mental health/substance related	4/5 strong matches. One image (drug use + cigarette) was slightly ambiguous	Excellent layout, responsive, theming applied correctly
OpenAI CLIP	2	16	Female	Very good — covers physical health, mental health, reproductive health, and safety	4/5 accurate or acceptable matches. One is mismatched (HIV image = Hepatitis)	Consistent layout, pink theming for gender applied correctly
OpenAI CLIP	3	59	Male	Good — includes infection, cancer, alcohol, and general screening	3/4 appropriate matches; 1 mismatch (HIV tip image = Hepatitis test label)	Clean layout, Bootstrap cards consistent, male theming color applied
OpenAI CLIP	4	40	Female	No health tips displayed — major content generation failure	N/A	Layout is visually clean, header and badge are styled appropriately

Table 4.3: *Evaluation of OpenAI CLIP as Replacement for BLIP*

Overall, the results indicate that while OpenAI CLIP is capable of producing competitive image-caption alignments, BLIP offers more reliable and semantically grounded results in this context. BLIP's text generation approach—conditioned directly on image features—proves better suited for precise healthcare image labeling, which is particularly important for sensitive or medical topics. Given the importance of image–tip coherence for user trust and readability, BLIP remains the preferred choice for the vision-language component of this system.

4.1.5 Output Evaluation

To objectively assess the overall performance of each model configuration, a scoring rubric was applied across three core dimensions; Tip Diversity, Image Match Accuracy, and Visual Quality. Each dimension was scored on a scale from 1 to 5, where 5 represents excellent performance and 1 indicates major failure or inaccuracy. The average of these three criteria represents the final output score for each model.

Model Configuration	Tip Diversity (/5)	Image Match Accuracy (/5)	Visual Quality (/5)	Final Score (/5)
CLIP + DeepSeek r1:7b	3.7	4.0	4.5	**4.07**
BLIP + DeepSeek r1:7b	4.0	3.3	4.3	**3.87**
BLIP + Gemma 3	4.3	2.7	4.2	**4.07**

Table 4.4: *Output Evaluation Scores Across Model Configurations*

The results in Table 4.4 offer several important observations:

- **Tip Diversity:** All three configurations scored highly in this area, with BLIP + Gemma 3 slightly outperforming the others by consistently generating a broader and more personalized set of tips.

- **Image Match Accuracy:** CLIP + DeepSeek r1:7b led in this category. Its strong semantic alignment between health tips and image captions likely stems from CLIP's dual-encoder architecture, which excels at embedding cross-modal similarity. BLIP-based configurations had more mismatches or ambiguous image connections.

- **Visual Quality:** All models produced well-structured HTML with consistent Bootstrap styling. CLIP + DeepSeek performed slightly better in terms of layout precision and theming consistency, but the differences were minor.

While both CLIP + DeepSeek and BLIP + Gemma 3 achieved the same final score (4.07), they did so through different strengths. The former offers stronger image alignment and visual polish, while the latter delivers enhanced topic relevance and diversity. This indicates that the optimal pairing depends on whether the design objective emphasizes semantic alignment or content richness.

4.2 Usability Evaluation

To complement the technical tests, a user-focused evaluation was conducted to assess the perceived effectiveness and satisfaction of the system. This section presents questionnaire results comparing the static and personalised prototypes from the perspective of end users.

4.2.1 Sampling Strategy

A total of ten participants were recruited using convenience sampling to evaluate the usability of both the static and LLM-generated prototypes. The sample was intentionally diverse, allowing for a meaningful analysis of how users with different demographic backgrounds perceive and interact with personalised web content.

The age of participants ranged from 16 to 63 years, representing a wide cross-section of user demographics. This distribution included younger users, middle-aged adults, and older users, thus providing insights across generational lines. Gender was relatively balanced, with six participants identifying as female and four as male.

Participants also reported varying levels of experience with health-related websites. Seven users had prior experience navigating digital health platforms, while the remaining three had little to no previous interaction with such systems. This variation was important in assessing both the accessibility of the prototypes for new users and the satisfaction levels of those familiar with health-based digital content.

By including a balanced sample in terms of age, gender, and health website familiarity, the evaluation ensured a more comprehensive and equitable understanding of the system's effectiveness in delivering personalised content. This diversity is particularly critical when assessing the performance of adaptive systems designed to tailor experiences based on user characteristics.

4.2.2 Static Prototype Feedback

Participants were first asked to evaluate the static version of the prototype, which presented generic health information without any personalisation based on user characteristics. The purpose of this assessment was to establish a baseline for usability, content relevance, and overall user satisfaction.

Quantitative responses revealed a moderately positive reception. On average, participants rated their overall experience using the static prototype at 3.9 out of 5, indicating general satisfaction with the platform's usability and layout. However, this score also suggests that the experience was not optimal for all users, particularly when compared to the more personalised alternative introduced later.

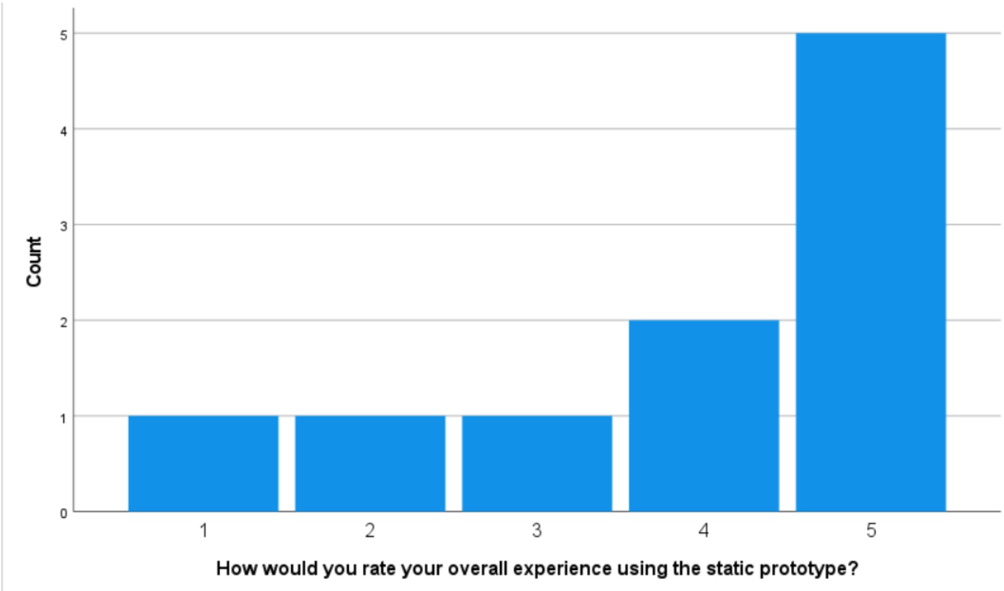

Figure 4.8: *Overall Experience Scores for Static Prototype*

In terms of content relevance, the static prototype received an average score of 4.3 out of 5. This indicates that participants found the health tips generally appropriate and useful, even without customisation. However, because the tips were not tailored to user age or gender, it is likely that some responses reflect a more passive or generic engagement with the material.

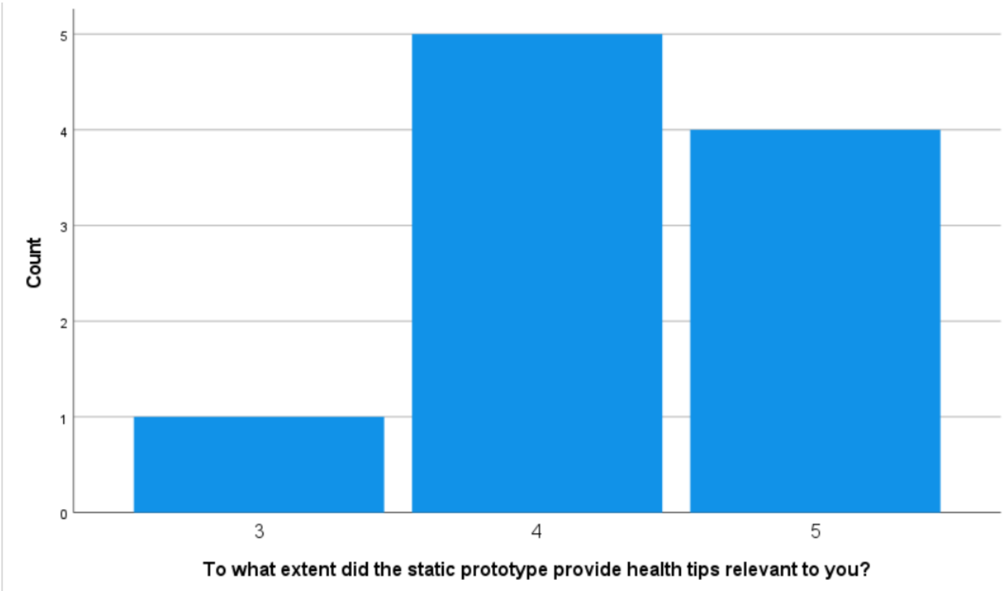

Figure 4.9: *Health Tip Relevancy Scores for Static Prototype.*

Qualitative feedback further highlights some areas for improvement. Several participants noted limitations in the static version's visual presentation and level of detail, with recurring suggestions related to increasing font size, adding colour, and enhancing the clarity of the health advice.

Overall, the static prototype performed adequately as a control condition, providing a functional and informative experience. However, its limitations in personal engagement and content specificity underscore the potential value of adaptive systems that respond to user characteristics.

4.2.3 LLM-Generated Prototype Feedback

Following the evaluation of the static version, participants were asked to assess the LLM-generated prototype, which dynamically tailored health tips and visual content based on user-provided demographic characteristics. The goal was to evaluate how effectively this personalised system enhanced the user experience compared to its static counterpart.

Quantitative results indicated a generally higher level of satisfaction with the LLM-generated version. The average rating for overall experience was 4.1 out of 5, an improvement over the static prototype's 3.9. Participants also gave high scores for the relevance of health tips, with an average of 4, and for the extent to which the content felt tailored to them, which received an average score of 4.4.

This high score is supported by participants who commented on the visual design and personalisation. For example, P7 described it as 'simple and eye-catching, with straightforward statements', while P10 explained that they 'appreciated the personalised articles and images'. Additionally, users rated their satisfaction with the system's personalisation at 4.1 out of 5, indicating broad approval of the adaptive content delivery approach.

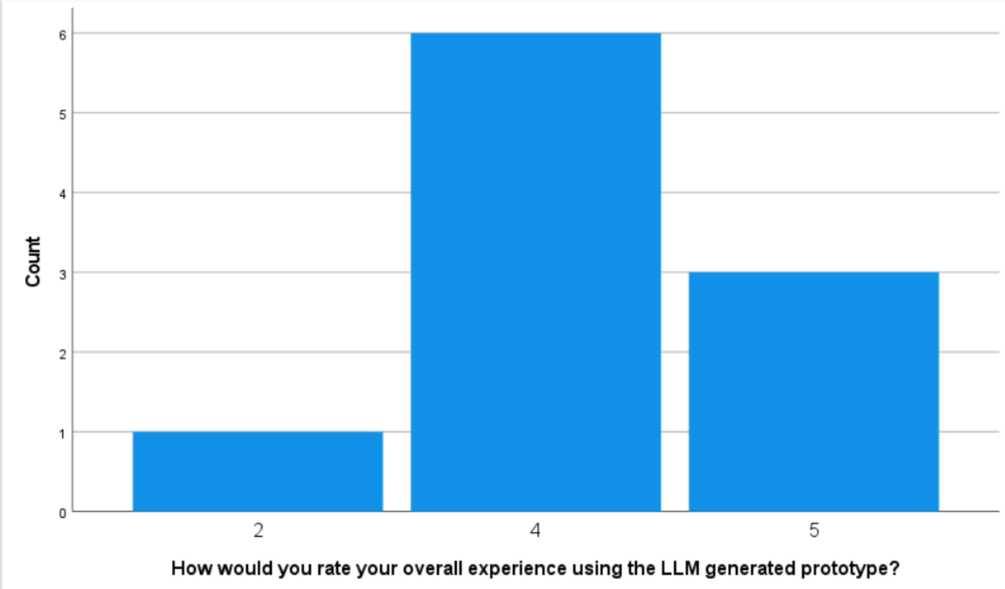

Figure 4.10: *Overall Experience Scores for LLM-Generated Prototype.*

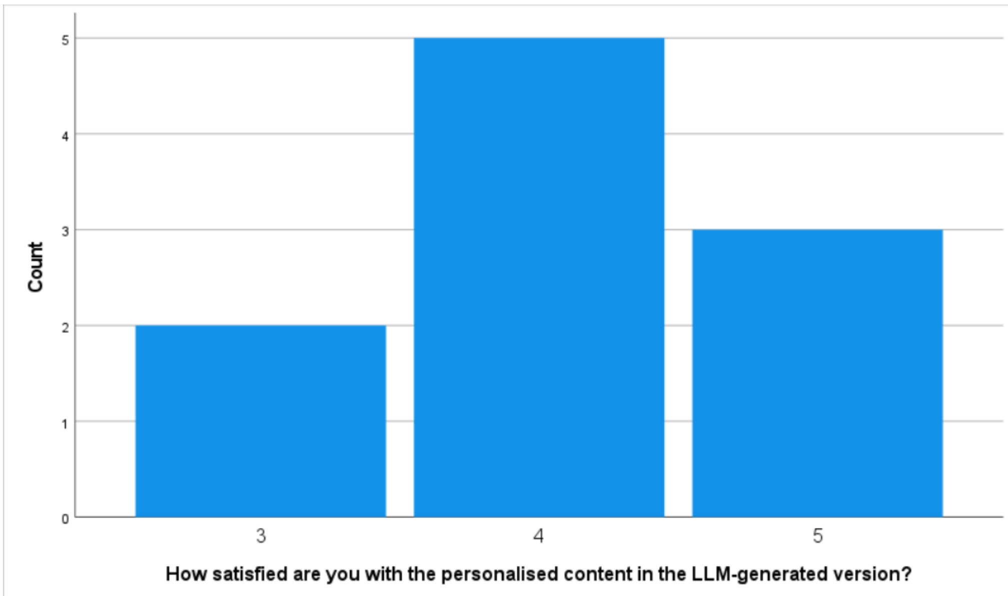

Figure 4.11: *Satisfaction Scores for LLM-Generated Prototype.*

Qualitative feedback explained these results. Many participants highlighted the prototype's engaging design and perceived alignment with their personal needs. Positive remarks centred on the presence of age-appropriate tips, visual appeal, and the intuitive layout of the personalised version. One participant noted, "I appreciate the personalised articles and images, even the colours of the web page," reflecting the value of contextual relevance.

Theme	**Summary of Participant Feedback**
Visual Design and Colour	Participants appreciated the use of colour, especially in the header. Several noted that the design was visually appealing and eye-catching (P1, P3, P7, P10).
Image Relevance	Multiple responses mentioned that images matched the tips well or were mostly relevant (P1, P5, P10).
Tip Relevance and Clarity	Health advice was described as straightforward, personalised, and easy to understand (P2, P6, P7, P10).
Content Variety	Participants valued the variety in the types of health tips provided (P4).
Suggestions for Improvement	Some participants offered enhancement ideas within their praise, such as using topic-based boxes or adding links to appointments (P8, P9).

Table 4.5: *Summary of open-ended feedback regarding the LLM-generated prototype*

However, some constructive feedback also emerged. A few users pointed out that certain images did not always match the accompanying health tips, which at times detracted from the perceived cohesion of the content. Others suggested improvements to layout spacing or text clarity. These issues point to areas where the vision-language pairing and visual presentation could be refined in future iterations.

Theme	**Summary of Participant Feedback**
Content Detail and Relevance	Several participants felt that the health tips were too brief or lacked personal relevance to their specific situation (P1, P4, P9, P10).
Layout and Organisation	Suggestions included improving the layout of health tips and placing related advice closer together (P2, P5).
Image–Tip Matching	One participant noted a mismatch between some tips and the accompanying images (P3).
Interactive Features	A few participants suggested enabling users to click on tips or explore related content further (P7).

Table 4.6: *Participant feedback on areas for improvement in the LLM-generated prototype*

Overall, the LLM-generated prototype was received more favourably than the static version across all evaluated criteria. The results suggest that users not only appreciated the tailored content but also found the personalised interface more engaging and informative. These findings support the hypothesis that adaptive systems driven by LLMs can meaningfully enhance user experience in web-based health communication.

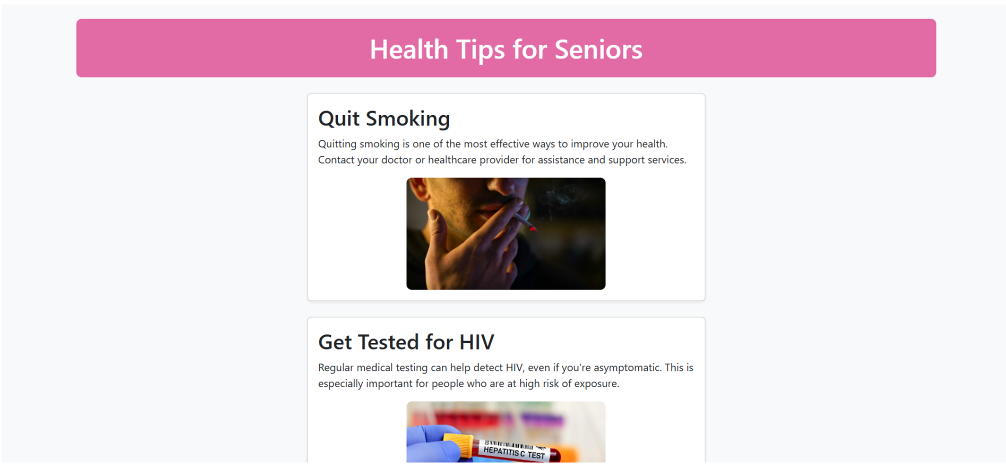

Figure 4.12: *Sample generation for 60 year old female.*

To further illustrate the output users interacted with, Figure 4.12 presents a sample page generated for a participant who selected Age: 60 and Gender: Female. The health tips focus on senior-relevant advice, such as smoking cessation and HIV screening, while the colour theme reflects gender-based styling preferences. This visual example demonstrates the system's ability to adapt both content and layout based on user characteristics, aligning with participants' feedback on relevance and personalisation.

4.2.4 Comparative Preference and Perceived Value

To evaluate the overall impact of personalisation and user satisfaction, participants were asked to indicate which version of the system they preferred, whether they felt the personalised content improved their experience, and whether they would recommend the system to others.

User characteristics were split evenly across the three available choices. Three participants preferred the LLM-generated prototype, three preferred the static version, and the remaining four reported no clear preference. This distribution suggests that while the LLM-generated version introduced novel benefits, the static version was still appreciated by some users, possibly due to its simplicity or familiarity. Those who favoured the LLM-based system often cited improved relevance and engaging presentation, with one user commenting, "The information was more tailored to me." On the other hand, participants who preferred the static version noted its clarity and directness.

Theme	Summary of Participant Feedback
Personalisation and Relevance	Participants appreciated that the content felt tailored to their characteristics, and noted that personalised delivery enhanced the experience (P1, P3).
Preference for Personalised Approach	One participant preferred the personalised version despite minor content misalignment, valuing the concept of adaptive delivery (P10).

Table 4.7: *Reasons participants preferred the LLM-generated prototype*

When asked whether the personalisation improved their experience, a more definitive trend emerged: nine out of ten participants answered yes, indicating that tailored content generally enhanced their interaction with the system, even among those who did not explicitly prefer the LLM version. This reinforces the idea that personalisation can add value independently of overall preference.

Theme	Summary of Participant Feedback
Improved Relevance	Participants noted that personalised content made the prototype more relevant and aligned with their age or needs (P1, P2, P3, P4, P7, P10).
Time-Saving and Usefulness	One participant pointed out that personalisation reduces time spent searching for relevant information (P5).
Motivation and Health Awareness	Some responses indicated that personalised tips enhanced motivation or awareness of healthy living (P6).
Clarity and Understanding	Personalised content helped improve comprehension and ease of use (P8).
Potential Future Use	One participant stated they may find such personalisation useful in the future, though they do not currently require it (P9).

Table 4.8: *Participant views on the impact of personalised content*

Participants were also asked whether they would recommend the system to others. Responses were largely positive, with six out of ten participants answering with ”Definitely Yes”. This suggests that despite some split in version preference, the system was broadly perceived as beneficial and effective in delivering useful health content.

In summary, although preferences for the two prototypes were evenly distributed, the perceived benefit of personalisation was strongly supported. This distinction highlights that while not all users may prefer a personalised interface outright, many still recognise its added value in delivering relevant and engaging content.

4.3 Summary of Key Findings

The usability evaluation yielded several key findings that reinforce the overall aim of the study: to assess whether adaptive, LLM-generated content enhances the user experience when compared to a static web prototype.

Firstly, the LLM-generated prototype consistently outperformed the static version across core usability metrics, including overall user experience, relevance of content, and perceived personalisation. Quantitative data indicated higher average ratings for the LLM version in all categories, with participants rating their satisfaction with the system’s tailored advice particularly favourably. This supports the hypothesis that even basic demographic tailoring can result in a more engaging and informative experience for users.

Secondly, user feedback revealed both benefits and limitations of the personalised system. Many participants appreciated the personalised tips, the visual layout, and the age-relevant content. However, some noted inconsistencies in the alignment between tips and images, and a few commented on layout clarity and content density. These insights point to the need for further refinement of the vision-language pairing process and the overall UI/UX design.

Thirdly, user characteristics for each prototype were mixed, with participants evenly divided across preferences for the static version, the LLM version, or no clear preference. Despite this, a large majority agreed that personalisation improved their experience, and nearly all participants stated they would recommend the system. This indicates that even when users did not explicitly prefer the LLM version, they still recognised the value it added in terms of relevance and personal connection.

Finally, the diversity of the participant group, which included a balanced gender mix and a wide age range, contributed to the robustness of these findings. Feedback from both experienced and inexperienced users of health websites suggests that personalised content delivery has broad applicability and potential across user groups.

In summary, the findings demonstrate that adaptive content generated by LLMs can significantly enhance user experience in web-based health communication — particularly when designed with inclusivity and user diversity in mind. The results also highlight areas for improvement, particularly in terms of visual-content alignment and layout polish, which will inform future iterations of the system.

4.4 Comparative Discussion with Related Studies

The findings from the previous chapter reveal clear trends in user preference for personalised content, alongside technical strengths and limitations. This chapter interprets these results in the context of the original research questions and theoretical framework.

These results demonstrate alignment with key findings from existing literature on personalisation, usability, and user-centred design. Both system-level evaluations and user feedback support the notion that adaptive content—especially when generated based on user characteristics can enhance the relevance and overall perceived value of web-based health information.

For instance, the high average ratings for content relevance and layout quality in the LLM-generated prototype are consistent with findings by Dianat et al. [11], who emphasised the importance of structural design elements such as layout and navigation in user satisfaction. Similarly, participants in this study cited improved organisation and personal alignment as reasons for preferring aspects of the personalised version [11].

Lee and Koubek [30] also found that structured layouts and clarity of content presentation are key drivers of usability. In the current study, several participants suggested improvements related to content grouping and layout refinement, reflecting similar concerns about structured interaction design.

The positive emotional responses to personalised content—reported by most participants—resonate with the work of Partala and Kallinen [3], who linked user satisfaction to experiences that support autonomy and competence. The LLM prototype, through tailored tips and age-specific recommendations, may have fostered these emotions by making users feel more understood and supported.

Additionally, Skadberg and Kimmel [4] introduced the concept of “flow” in web experiences, identifying interactivity and content relevance as core factors. Although the current prototype did not include dynamic interaction beyond content presentation, the relevance of the personalised tips contributed to a more engaging experience, as supported by qualitative feedback.

Vu et al. [31] further emphasised the need to accommodate users with different cognitive and physical abilities through inclusive web design. The findings of this study—particularly the positive feedback from participants across age groups—suggest that basic demographic tailoring may be an effective first step toward that inclusivity, although more complex adaptivity could further improve outcomes.

Overall, the study supports a broader academic consensus that personalisation and user-centred design improve user experience, especially in informational domains like healthcare. However, limitations such as minor image–tip mismatches and layout refinement needs mirror challenges identified in the literature, highlighting areas for further iteration and testing.

Chapter 5: Conclusions and Recommendations

5.1 Main Conclusions

This study sought to develop and evaluate a proof-of-concept adaptive web content delivery system that leverages LLMs to generate personalised health information. Specifically, the system aimed to tailor content dynamically based on user characteristics such as age and gender using DeepSeek r1:7b for language generation and BLIP for image matching.

The results indicate that the hypothesis holds true. However, certain observations and suggestions for improvement mentioned in the results of this study require consideration. For example, while participants appreciated the personalised tips, some noted that the layout could be better organised, and certain tips were too brief or lacked actionable depth. There were also occasional mismatches between tips and images, which affected coherence in some cases. Additionally, although overall feedback was positive, user preferences were not unanimous as three out of the ten participants favoured the static version due to its simplicity and familiarity. These findings highlight areas where usability and content alignment can be refined.

Participants rated the LLM-generated prototype higher in terms of relevance and overall satisfaction, and the majority agreed that personalisation improved their experience. This can be seen in Figures 4.10 and 4.11.

These results directly address the research questions defined in this dissertation. Research Question One explored whether the use of LLMs could improve the personal relevance of health content. This was supported by the quantitative ratings and qualitative feedback from participants shown in Table 4.5. Research Question Two focused on usability, comparing adaptive and static designs. While personalisation was valued, preferences were mixed, indicating that familiarity and layout simplicity also influenced usability outcomes. Research Question Three asked whether the system would function consistently across repeated generations. System-level tests confirmed the pipeline's stability, although image-caption matching required further refinement.

5.2 Limitations and Suggestions for Improvement

Despite the promising results observed, several limitations affected the scope and robustness of this study. One of the key limitations concerned the image-matching process, which was handled using the BLIP vision-language model. Although generally effective, this pipeline occasionally failed to retrieve contextually accurate or relevant images. Health Tip #5, in particular, consistently triggered mismatches or resulted in missing images. These issues underscore the need for more reliable fallback logic—such as confidence-based filtering, default placeholder images, or secondary matching attempts—to ensure visual consistency and reduce user confusion.

Another limitation lies in the narrow personalisation criteria used. The system only considered two demographic inputs—age and gender—when generating content. While these variables enabled basic segmentation and produced reasonably relevant health advice, they do not fully capture the complexity of real-world user needs. Factors such as pre-existing medical conditions, health literacy, geographic location, or personal health goals were not accounted for. A richer profiling mechanism could improve the specificity and applicability of the advice generated.

In terms of sample size, the user evaluation was limited to ten participants. While suitable for exploratory usability testing, this number restricts the statistical generalisability of the findings. Furthermore, the participant pool skewed toward younger, digitally literate users, meaning the results may not accurately reflect how older adults or those with limited technical experience would interact with the system. Future research should aim to include more diverse and representative user groups to enhance external validity.

The study also faced deployment constraints, as the prototype was not hosted online but instead generated and displayed as a locally saved HTML file. This approach, while practical for testing purposes, limited the realism of the user experience and made it harder to evaluate performance factors such as load time, responsiveness, or device compatibility in a live setting. These issues highlight a technical limitation in simulating real-world usage conditions.

Lastly, minor performance inconsistencies were observed in content generation time and layout alignment, though these did not significantly hinder usability. Addressing these issues will be critical for improving system responsiveness and polish in future iterations.

Building on these constraints, several improvements and research opportunities are suggested below.

5.3 Recommendations for Future Research

Future research should explore expanding the system's personalisation capabilities by integrating a more comprehensive set of user characteristics beyond age and gender. These may include medical history, health goals, language proficiency, accessibility needs, and even location-based data to provide more context-aware advice. Figure 5.1 outlines a proposed development pipeline for improving future versions of the system.

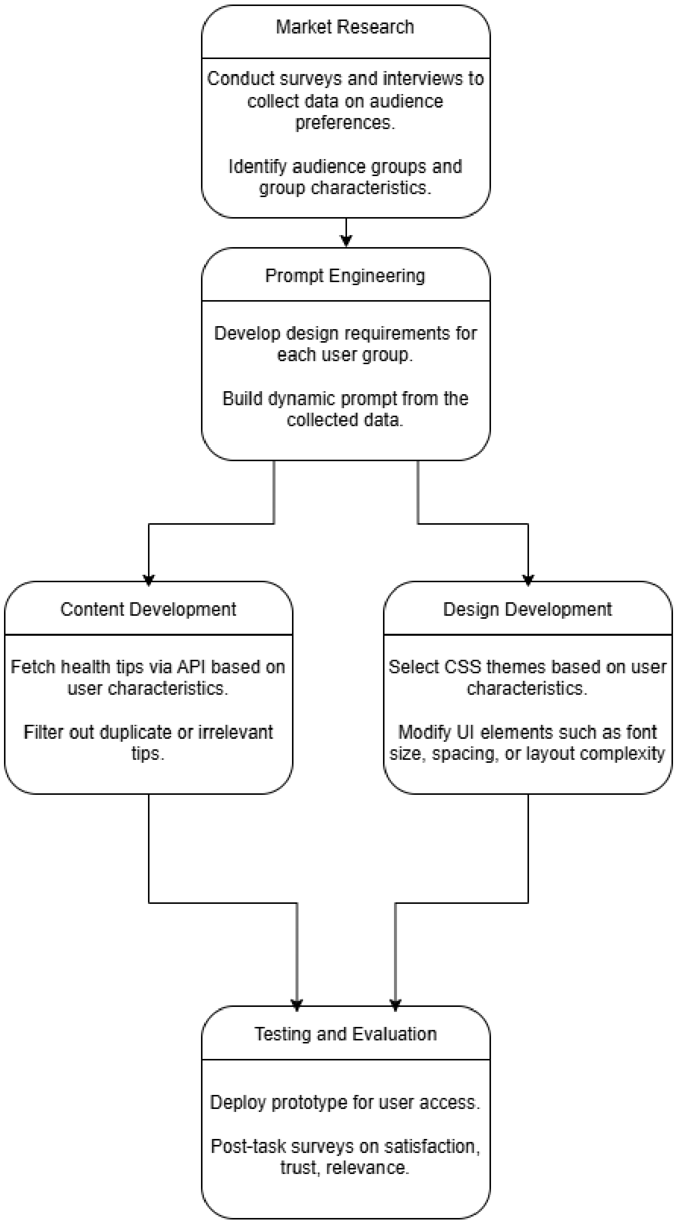

Figure 5.1: *Proposition for a development pipeline of personalised design and content of websites.*

In particular, applying Gardner's theory of multiple intelligences—which distinguishes between visual-spatial, linguistic-verbal, bodily-kinesthetic, interpersonal, intrapersonal, and other learning styles—could enable the system to adapt not only the content but also the format of delivery to suit individual cognitive preferences [36]. For example, visual learners might benefit more from infographicbased advice, while verbal learners may prefer detailed textual descriptions.

To support this, future iterations could allow users to self-identify or infer these preferences through onboarding questions or usage behaviour, enabling deeper personalisation. A real-time feedback loop could also be introduced, allowing users to rate or customise content, which in turn would train the system to better align with their preferences over time.

On the technical side, algorithmic enhancements should be considered to improve performance, relevance, and robustness. For example, the image-caption matching pipeline could be upgraded by combining multiple vision-language models or integrating semantic similarity filters to reduce visual mismatches. Deploying confidence-based logic, where low-confidence matches are replaced with generic but safe alternatives, would further increase reliability. Additionally, incorporating content diversity and filtering logic could prevent repetition and improve content freshness across sessions.

Deployment improvements are also essential for broader applicability. Hosting the prototype online through a scalable cloud-based backend would allow for cross-device accessibility and real-time content generation, enabling performance benchmarking under more realistic usage conditions. This would also support future enhancements such as user authentication, persistent preferences, and device-aware design.

Finally, broader-scale user testing involving diverse demographics and longer-term engagement would offer stronger evidence for the system's practical effectiveness and scalability. Future iterations could benefit from incorporating user modelling inspired by educational theories, such as Gardner's Theory of Multiple Intelligences, which recognises diverse cognitive styles including visual, verbal, and kinesthetic learning preferences [36]. These future directions can help transform this prototype into a reliable and inclusive tool for delivering adaptive, user-aware digital content in health and beyond.

These directions would help transform the current prototype from a proof-of-concept into a mature, user-centric platform capable of delivering high-quality adaptive content across multiple domains.

As AI systems continue to shape the digital landscape, approaches like the one explored in this study offer a pathway toward more human-centred and context-aware applications.

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Chapter A: Appendix

A.1 Questionnaire Consent Form

Consent Form

Dear Participant, I am currently reading for a bachelor's degree (Hons) at the Malta College of Arts, Science and Technology (MCAST). The title of my research is "Adaptive Web Content Delivery Using Large Language Models: Personalizing User Experience Based on User Characteristics." This study explores how generative Artificial Intelligence (AI) models can be used to adapt digital content to better suit users based on attributes such as age, gender, and preferences. The aim is to improve accessibility, engagement, and overall user experience across diverse user groups.

You are being invited to take part in this research study. If you agree to participate, you will be asked to test a web content delivery prototype that was developed using AI and compare it with another prototype created without AI assistance. Following this, you will be asked to complete a short questionnaire about your experience with both versions. The session will conclude with a brief 10-minute interview. By participating, you will contribute to the development of guidelines and insights into the ethical and effective use of AI for personalizing web content, particularly through the lens of human-computer interaction and user experience design. If you choose to participate, you will be provided with this information sheet and asked to sign a corresponding consent form.

There are no foreseeable disadvantages or risks associated with taking part in this study, and all ethical guidelines set by MCAST will be strictly followed. Any information collected during the research will be kept confidential and only accessible to the researcher. Your responses will remain anonymous and will not be linked to you or your organization in any published material.

The information gathered will be used exclusively for research purposes. The interview will focus solely on your experiences and perspectives regarding the use of AI for web personalization. You will not be asked questions about personal beliefs, teaching practices, or any sensitive work-related matters. Any potentially identifying information will be excluded from the final analysis and publication. Data may be retained and used in anonymised form for the duration of the study and will be destroyed upon its completion. By agreeing to participate in this study, you consent to the collection, retention, and anonymised publication of your input as described.

If you have any concerns about any aspect of the way in which you have been approached or treated during this study, do not hesitate to contact the researcher on the following email address: leon.sacco.f32226@mcast.edu.mt. Moreover, if you would like more information about the research study before you decide whether to participate or not, feel free to contact the researcher on the above-mentioned email address. Thank you for your cooperation and interest shown in this research.

Figure A.1: *Consent form present at the start of the questionnaire.*

A.2 Questionnaire Questions

6/3/25, 10:38 AM Dissertation Qualitative Questionnaire

1. I confirm that I have read and understand the Information Letter for the above study * and have had the opportunity to ask questions.

   I also understand that my/my charge's participation is voluntary and that I/my charge am/are free to withdraw at any time without giving any reason.

   Mark only one oval.

   - Yes
   - No

Participant Preferences

Please answer the following based on your personal preferences.

2. What is your age? *

   ____

3. What is your gender? *

   Mark only one oval.

   - Male
   - Female

4. Do you have experience using health-related websites? *

   Mark only one oval.

   - Yes
   - No

https://docs.google.com/forms/d/1L-CtyVTVQsAQnFSFASN8U42FPIW7e5rogquQdLlhtI/edit 3/8

Figure A.2: *First page of questionnaire.*

6/3/25, 10:38 AM Dissertation Qualitative Questionnaire

Static Prototype Questions

Please make sure you have viewed the static prototype using the link below before answering this section.

https://leonsacco21.github.io/dissertation-static/

5. How would you rate your overall experience using the static prototype? *

Mark only one oval.

	1	2	3	4	5	
Very	◯	◯	◯	◯	◯	Excellent

6. To what extent did the static prototype provide health tips relevant to you? *

Mark only one oval.

	1	2	3	4	5	
Not	◯	◯	◯	◯	◯	Extremely Relevant

7. What would you improve in this version? *

https://docs.google.com/forms/d/1L-CtyVTVQsAQnFSFASh8U42FPIW7e5rogquQdLhII/edit 4/8

Figure A.3: *Second page of questionnaire.*

6/3/25, 10:38 AM Dissertation Qualitative Questionnaire

LLM-Generated Prototype Questions

Please make sure you have viewed the LLM-generated prototype using the link below before answering this section.

https://leonsacco21.github.io/dissertation-custom/

8. How would you rate your overall experience using the LLM generated prototype? *

Mark only one oval.

	1	2	3	4	5	
Very	◯	◯	◯	◯	◯	Excellent

9. To what extent did the LLM-generated prototype provide health tips relevant to you? *

Mark only one oval.

	1	2	3	4	5	
Not	◯	◯	◯	◯	◯	Extremely Relevant

10. Did the LLM-generated prototype tailor the content to your age and gender effectively? *

Mark only one oval.

	1	2	3	4	5	
Not	◯	◯	◯	◯	◯	Completely

https://docs.google.com/forms/d/1L-CtyVTVQsAQnFSFASN8U42FPIW7e5rogquQdLIhtI/edit 5/8

Figure A.4: *Third page of questionnaire.*

6/3/25, 10:38 AM Dissertation Qualitative Questionnaire

11. How satisfied are you with the personalised content in the LLM-generated version? *

Mark only one oval.

	1	2	3	4	5	
Very	◯	◯	◯	◯	◯	Very Satisfied

12. What did you like most about the LLM-generated prototype? *

13. What aspects of the LLM-generated prototype could be improved? *

Comparative Reflection

This section asks for your overall opinion of both prototypes and which version you preferred.

https://docs.google.com/forms/d/1L-CtyVTVQsAQnFSFASh|8U42FPIW7e5rogquQdLIhtI/edit 6/8

Figure A.5: *Fourth page of questionnaire.*

6/3/25, 10:38 AM Dissertation Qualitative Questionnaire

14. Which version did you prefer overall? *

Mark only one oval.

- ( ) LLM-Generated Prototype
- ( ) Static Prototype
- ( ) No Preference

15. In a few words, why did you prefer the version selected above? *

16. Do you think personalised content made a difference in your experience? And why? *

https://docs.google.com/forms/d/1L-CtyVTVQsAQnFSFASN8U42FPIW7e5rogquQdLlhIl/edit 7/8

Figure A.6: *Fifth page of questionnaire.*

6/3/25, 10:38 AM Dissertation Qualitative Questionnaire

17. Would you recommend a personalised healthcare tips webpage like this to others? *

Mark only one oval.

- ( ) Definitely Not
- ( ) Probably Not
- ( ) Not Sure
- ( ) Probably Yes
- ( ) Definitely Yes

Final Thoughts and Submission

18. Do you have any other suggestions or feedback?

____

____

____

____

____

This content is neither created nor endorsed by Google.

Google Forms

https://docs.google.com/forms/d/1L-CtyVTVQsAQnFSFASh8U42FPIW7e5rogquQdLIhtI/edit 8/8

Figure A.7: *Sixth page of questionnaire.*

A.3 Sample Generations

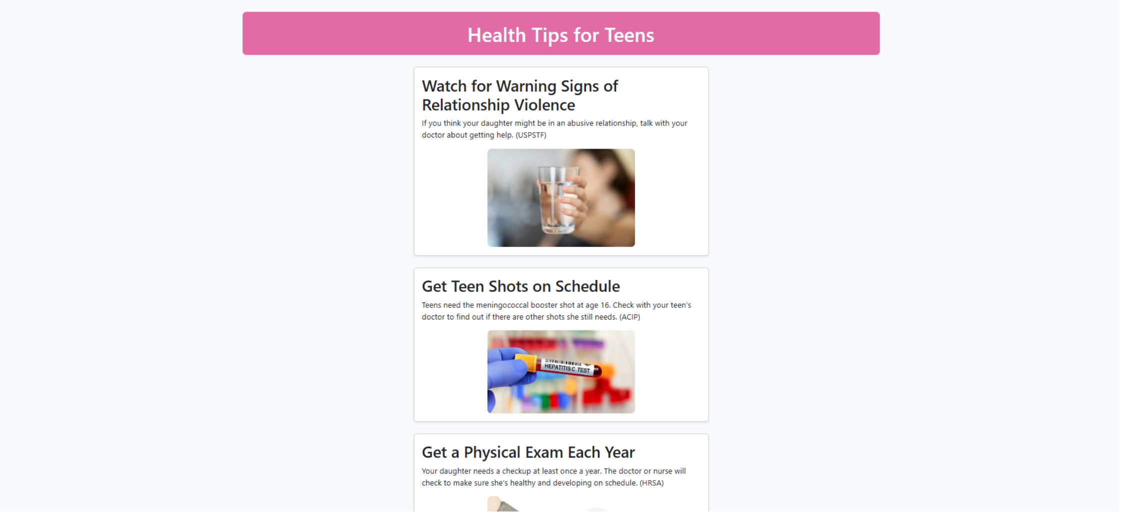

Figure A.8: *Generated sample for a 16-year-old female user.*

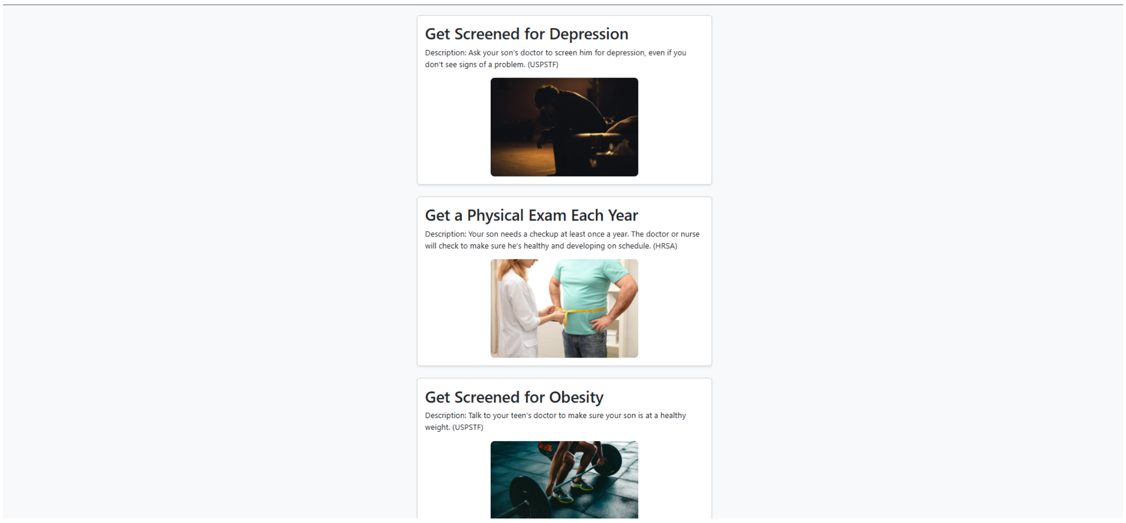

Figure A.9: *Generated sample for a 17-year-old male user.*

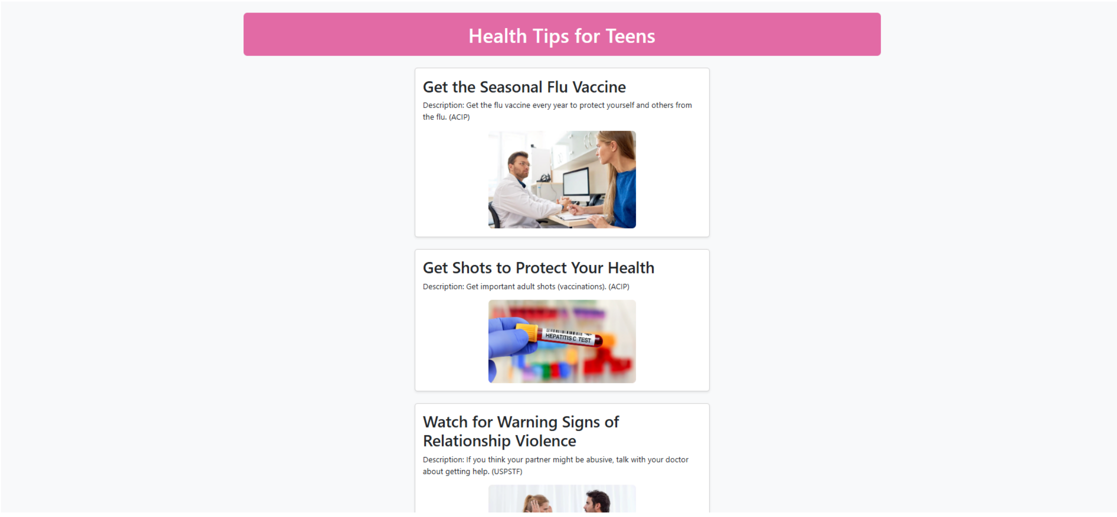

Figure A.10: *Generated sample for a 19-year-old female user.*

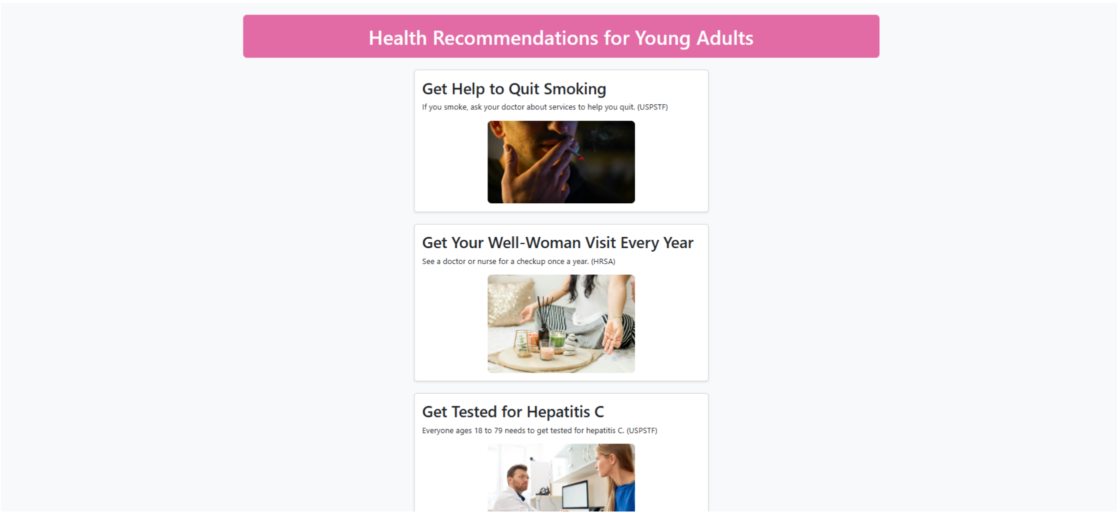

Figure A.11: *Generated sample for a 40-year-old female user.*

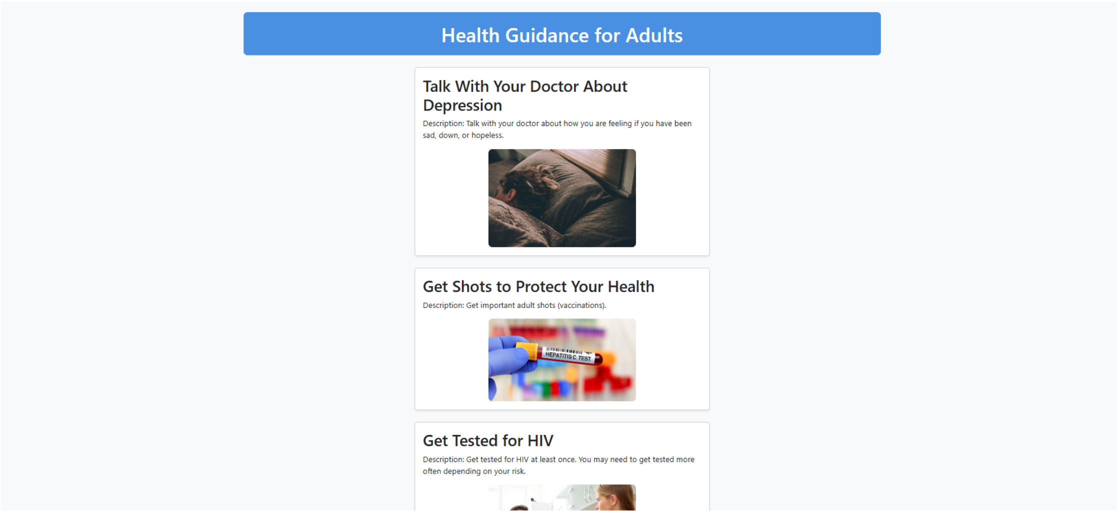

Figure A.12: *Generated sample for a 52-year-old male user.*

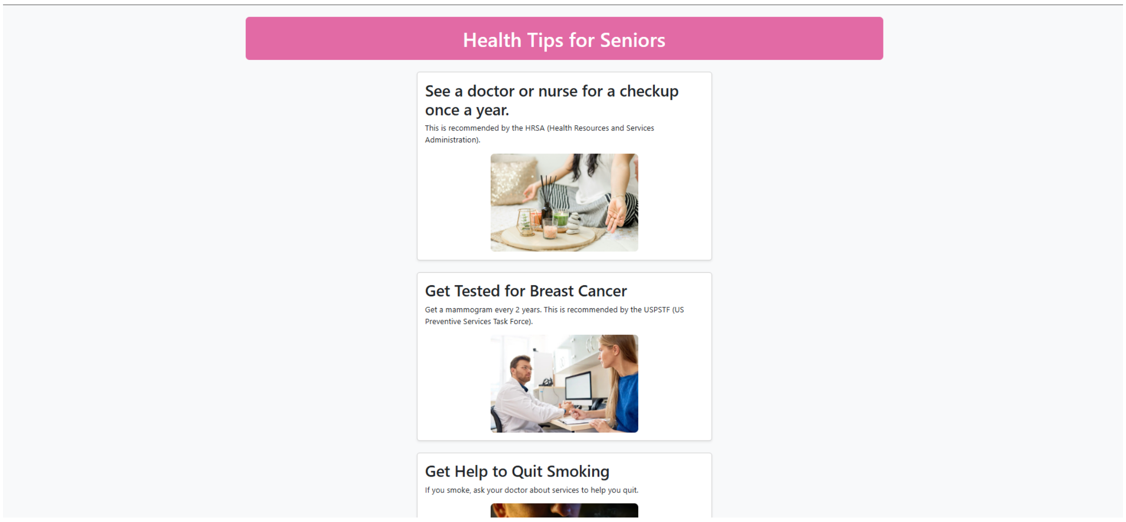

Figure A.13: *Generated sample for a 63-year-old female user.*

Chapter B: Sample Code

```python
import os
import re
import torch
import json
import random
import requests
import subprocess
import gradio as gr
from PIL import Image
from transformers import BlipProcessor, BlipForConditionalGeneration
from sentence_transformers import SentenceTransformer, util
import webbrowser

# --- Load Models ---
blip_processor = BlipProcessor.from_pretrained("Salesforce/blip-image-captioning-base")
blip_model =
    BlipForConditionalGeneration.from_pretrained("Salesforce/blip-image-captioning-base")
blip_model.to("cuda" if torch.cuda.is_available() else "cpu")

embedder = SentenceTransformer("all-MiniLM-L6-v2")

# --- Constants ---
IMAGE_DIR = "llava-images"
OUTPUT_HTML_FILE = "prototype-final.html"
OLLAMA_MODEL = "deepseek-r1:7b"

# --- Step 1: Generate Image Captions ---
def generate_image_captions():
    captions = {}
    for filename in os.listdir(IMAGE_DIR):
        if filename.lower().endswith((".jpg", ".jpeg", ".png")):
            path = os.path.join(IMAGE_DIR, filename)
            try:
                image = Image.open(path).convert("RGB")
                inputs = blip_processor(image,
                    return_tensors="pt").to(blip_model.device)
                out = blip_model.generate(**inputs)
                caption = blip_processor.decode(out[0], skip_special_tokens=True)
                captions[path] = caption
            except Exception as e:
                print(f"Skipping {filename}: {e}")
    return captions

# --- Step 2: Get Health Tips ---
def get_health_tips(age, gender):
    gender = gender.lower()
    url =
        f"https://odphp.health.gov/myhealthfinder/api/v3/myhealthfinder.json?age={age}&sex={gender}"

    try:
        response = requests.get(url)
        data = response.json()
        resources = data["Result"]["Resources"]["all"]["Resource"]
        tips = []
        for res in resources:
            title = res.get("MyHFTitle", "No Title")
            desc_html = res.get("MyHFDescription", "")
            desc = re.sub(r"<.*?>", "", desc_html).strip()
            tips.append((title, desc))
        return random.sample(tips, k=min(5, len(tips)))
    except Exception as e:
        print(f"API error: {e}")
```

```
        return []

# --- Step 3: Match Tips with Images ---
def match_tips_with_images(tips, image_captions):
    used = set()
    tip_texts = [f"{t[0]}. {t[1]}" for t in tips]
    tip_embeddings = embedder.encode(tip_texts, convert_to_tensor=True)

    image_texts = list(image_captions.values())
    image_paths = list(image_captions.keys())
    image_embeddings = embedder.encode(image_texts, convert_to_tensor=True)

    results = []

    for i, tip_emb in enumerate(tip_embeddings):
        sims = util.cos_sim(tip_emb, image_embeddings)[0]
        sorted_indices = torch.argsort(sims, descending=True)

        for idx in sorted_indices:
            img_path = image_paths[idx]
            if img_path not in used:
                used.add(img_path)
                results.append({
                    "title": tips[i][0],
                    "description": tips[i][1],
                    "image": img_path.replace("\\", "/")
                })
                break
    return results

def generate_page_title(age, gender):
    if age < 13:
        return "Health Suggestions for Children"
    elif 13 <= age <= 19:
        return "Health Tips for Teens"
    elif 20 <= age <= 40:
        return "Health Recommendations for Young Adults"
    elif 41 <= age <= 59:
        return "Health Guidance for Adults"
    else:
        return "Health Tips for Seniors"

# --- Step 4: Generate Prompt for LLM ---
def generate_prompt(matched_data, age, gender):
    # --- Design preferences based on user profile ---
    if gender.lower() == "male" and age < 30:
        design_note = "Use a modern, minimalistic layout with bold fonts and strong
            contrast."
    elif gender.lower() == "female" and age < 30:
        design_note = "Use an engaging layout with soft colors, rounded components,
            and clear structure."
    elif age >= 60:
        design_note = "Ensure accessibility with large fonts, high contrast, and
            simple layout."
    else:
        design_note = "Use a clean, readable layout with balanced colors and
            structured spacing."

    print(f"[INFO] Design note: {design_note}")

    page_title = generate_page_title(age, gender)

    prompt = (
        f"Generate a simple HTML page for a healthcare website titled
            '{page_title}'.\n"
        f"{design_note}\n"
        "- Add a main heading at the top of the page using <h1> that shows a related
            title.\n"
        "- For each health tip:\n"
        "  - Wrap the tip content in a <div style='margin: 20px 0;'> block.\n"
```

```
        "  - Inside the div, include:\n"
        "    - A heading using <h2> for the title.\n"
        "    - A <p> tag for the description.\n"
        "    - An <img> tag for the image using inline style 'width: 300px'.\n"
        "- Do not use JavaScript or external CSS frameworks.\n"
    )

    for item in matched_data:
        prompt += f"Title: {item['title']}\n"
        prompt += f"Description: {item['description']}\n"
        prompt += f"Image: {item['image']}\n\n"

    return prompt

# --- Step 5: Run Ollama Model Locally ---
def call_ollama(prompt):
    try:
        result = subprocess.run(
            ["ollama", "run", OLLAMA_MODEL],
            input=prompt.encode("utf-8"),
            stdout=subprocess.PIPE,
            stderr=subprocess.PIPE,
            timeout=300
        )
        return result.stdout.decode("utf-8")
    except Exception as e:
        print(f"Ollama error: {e}")
        return ""

# --- Clean Generated HTML from unwanted code ---
def extract_valid_html(content):
    start = content.lower().find("<!doctype html")
    end = content.lower().find("</html>") + len("</html>")
    if start != -1 and end != -1:
        return content[start:end]
    return content  # fallback: return everything if not well-formatted

def remove_javascript(content):
    return re.sub(r"<script.*?>.*?</script>", "", content, flags=re.DOTALL |
        re.IGNORECASE)

def resize_images_inline(content, width_px=300):
    return re.sub(r'<img([^>]+)>', rf'<img\1 style="width: {width_px}px;">', content,
        flags=re.IGNORECASE)

def insert_healthfinder_badge(content):
    badge_html = """
    <div class="text-center mt-5 pb-5">
        <a href="https://odphp.health.gov/myhealthfinder" title="MyHealthfinder">
            <img
                src="https://odphp.health.gov/themes/custom/healthfinder/images/MyHF.svg"
                alt="MyHealthfinder" style="max-width: 200px;" />
        </a>
    </div>
    """

    # Insert just before </body>
    if "</body>" in content:
        content = content.replace("</body>", badge_html + "\n</body>")
    else:
        content += badge_html

    return content

# --- Step 6: Save HTML ---
def save_html(content):
    cleaned_html = remove_javascript(content)
    resized_html = resize_images_inline(cleaned_html)
    with open(OUTPUT_HTML_FILE, "w", encoding="utf-8") as f:
        f.write(resized_html)
```

```
    return f"HTML page saved to: {OUTPUT_HTML_FILE}"

def apply_bootstrap_to_html(filepath):
    if not os.path.exists(filepath):
        print(f"[ERROR] File not found: {filepath}")
        return

    with open(filepath, "r", encoding="utf-8") as f:
        html = f.read()

    # Inject Bootstrap CDN
    bootstrap_link = (
        '<link
            href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.css"
            '
        'rel="stylesheet" integrity="sha384-..." crossorigin="anonymous">'
    )
    html = html.replace("<head>", f"<head>\n  {bootstrap_link}")

    # Wrap body content in a Bootstrap container
    html = re.sub(r"<body>", "<body>\n  <div class=\"container mt-4\">", html,
         flags=re.IGNORECASE)
    html = re.sub(r"</body>", "  </div>\n</body>", html, flags=re.IGNORECASE)

    # Add 'img-fluid' class to images
    html = re.sub(r"<img([^>]*?)>", r'<img\1 class="img-fluid">', html,
         flags=re.IGNORECASE)

    # Optional: add spacing classes to headers
    html = re.sub(r"<h1>", '<h1 class="text-center mb-4">', html)
    html = re.sub(r"<h2>", '<h2 class="mt-4">', html)

    # Save modified file
    with open(filepath, "w", encoding="utf-8") as f:
        f.write(html)

    print(f"[INFO] Bootstrap added to: {filepath}")

def clean_up_generated_html(filepath):
    if not os.path.exists(filepath):
        print(f"[ERROR] File not found: {filepath}")
        return

    with open(filepath, "r", encoding="utf-8") as f:
        html = f.read()

    # Fix malformed image style tags
    html = re.sub(
        r"<img([^>]*?)style=['\"]?[^>]*?['\"]?\s*style=['\"]?[^>]*?['\"]?",
        r'<img\1',
        html,
        flags=re.IGNORECASE,
    )

    # Fix any broken quote in style
    html = re.sub(
        r'style=[\'\"]([^\'\"]+)[\'\"]([^>]*)style=[\'\"][^\'\"]+[\'\"]',
        r'style="\1"\2',
        html
    )

    # Simplify image tags to use a single clean style
    html = re.sub(
        r'<img([^>]*?)>',
        r'<img\1 style="width: 300px;" class="img-fluid">',
        html
    )

    # Remove nested .container
    html = html.replace('<div class="container">', '')  # remove inner
```

```
    html = html.replace('</div>\n  </div>', '</div>')  # close only once

    # Optionally remove unused inline CSS block
    html = re.sub(r"<style>.*?</style>", "", html, flags=re.DOTALL)

    # Save cleaned result
    with open(filepath, "w", encoding="utf-8") as f:
        f.write(html)

    print(f"[INFO] Cleaned up and saved: {filepath}")

def apply_gender_theming(filepath, gender):
    if not os.path.exists(filepath):
        print(f"[ERROR] File not found: {filepath}")
        return

    with open(filepath, "r", encoding="utf-8") as f:
        html = f.read()

    # Determine color
    primary_color = "#4a90e2" if gender.lower() == "male" else "#e26aa5"

    # Inject a <style> block just before </head>
    style_block = f"""
    <style>
        body {{
            background-color: #f8f9fa;
        }}
        h1 {{
            background-color: {primary_color};
            color: white;
            padding: 20px;
            border-radius: 8px;
        }}
    </style>
    """

    html = html.replace("</head>", style_block + "\n</head>")

    with open(filepath, "w", encoding="utf-8") as f:
        f.write(html)

    print(f"[INFO] Gender theme applied for: {gender}")

def wrap_health_tips_in_cards(filepath, page_title="Health Tips"):
    if not os.path.exists(filepath):
        print(f"[ERROR] File not found: {filepath}")
        return

    with open(filepath, "r", encoding="utf-8") as f:
        html = f.read()

    # Find sequences of <h2>, <p>, and <img>
    tip_blocks = re.findall(
        r'(<h2[^>]*?>.*?</h2>\s*<p[^>]*?>.*?</p>\s*<img[^>]*?>)',
        html,
        flags=re.DOTALL | re.IGNORECASE
    )

    if not tip_blocks:
        print("[WARN] No tip blocks matched. Check HTML structure.")
        return

    # Extract existing badge (if it exists)
    badge_match = re.search(
        r'(<a href="https://odphp\.health\.gov/myhealthfinder".*?</a>)',
        html,
        flags=re.DOTALL | re.IGNORECASE
    )
    badge_html = f"""
```

```
    <div class="text-center mt-5">
      {badge_match.group(1)}
    </div>
    """ if badge_match else ""

    # Build Bootstrap cards
    card_html = ""
    for block in tip_blocks:
        title = re.search(r'<h2[^>]*?>(.*?)</h2>', block, re.DOTALL)
        desc = re.search(r'<p[^>]*?>(.*?)</p>', block, re.DOTALL)
        img = re.search(r'<img[^>]*?>', block, re.DOTALL)

        if title and desc and img:
            card = f"""
            <div class="card mb-4 p-3 shadow-sm" style="width: 100%; max-width:
                600px;">
              <h2 class="card-title">{title.group(1).strip()}</h2>
              <p class="card-text">{desc.group(1).strip()}</p>
              {img.group(0).replace('<img', '<img class="img-fluid rounded-3 mx-auto
                  d-block"')}
            </div>
            """
            card_html += card

    layout = f"""
    <div class="d-flex flex-column align-items-center">
      {card_html}
      {badge_html}
    </div>
    """

    # Replace <body> content without deleting the badge
    html = re.sub(
        r'(?is)<body.*?>.*?</body>',
        f'<body>\n  <div class="container mt-4">\n  <h1 class="text-center
            mb-4">{page_title}</h1>\n{layout}\n</div>\n</body>',
        html
    )

    with open(filepath, "w", encoding="utf-8") as f:
        f.write(html)

    print("[INFO] Health tips wrapped in Bootstrap cards with badge preserved.")

# --- Pipeline Triggered by Gradio ---
def gradio_callback(age, gender):
    print(f"\n[INFO] Starting pipeline for Age={age}, Gender={gender}")

    tips = get_health_tips(age, gender)
    if not tips:
        return "No health tips found."

    image_captions = generate_image_captions()
    matched_data = match_tips_with_images(tips, image_captions)

    prompt = generate_prompt(matched_data, age, gender)
    raw_output = call_ollama(prompt)
    html_output = extract_valid_html(raw_output)
    html_output = insert_healthfinder_badge(html_output)

    if "<html" in html_output.lower():
        page_title = generate_page_title(age, gender)

        msg = save_html(html_output)
        apply_bootstrap_to_html(OUTPUT_HTML_FILE)
        clean_up_generated_html(OUTPUT_HTML_FILE)
        apply_gender_theming(OUTPUT_HTML_FILE, gender)
        wrap_health_tips_in_cards(OUTPUT_HTML_FILE, page_title)

        # Open the generated HTML file in a new browser tab
```

```
        file_url = f"file://{os.path.abspath(OUTPUT_HTML_FILE)}"
        webbrowser.open_new_tab(file_url)

        return msg
    else:
        print(html_output)
        return "Failed to generate valid HTML from Ollama."

# --- Gradio UI ---
with gr.Blocks() as demo:
    gr.Markdown("## Generate Personalized Health Tips Page")

    with gr.Row():
        age_input = gr.Number(label="Age", minimum=0)
        gender_input = gr.Dropdown(choices=["Male", "Female"], label="Gender")

    submit_btn = gr.Button("Generate HTML Page")
    result_output = gr.Textbox(label="Status")

    submit_btn.click(fn=gradio_callback, inputs=[age_input, gender_input],
        outputs=result_output)

demo.launch(share=True)
```